

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

    #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
elif camp == "2025_Sep":
    mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7876
Nombre de lignes df clean	7876
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5513
Taille dataset test	2362
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :
id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

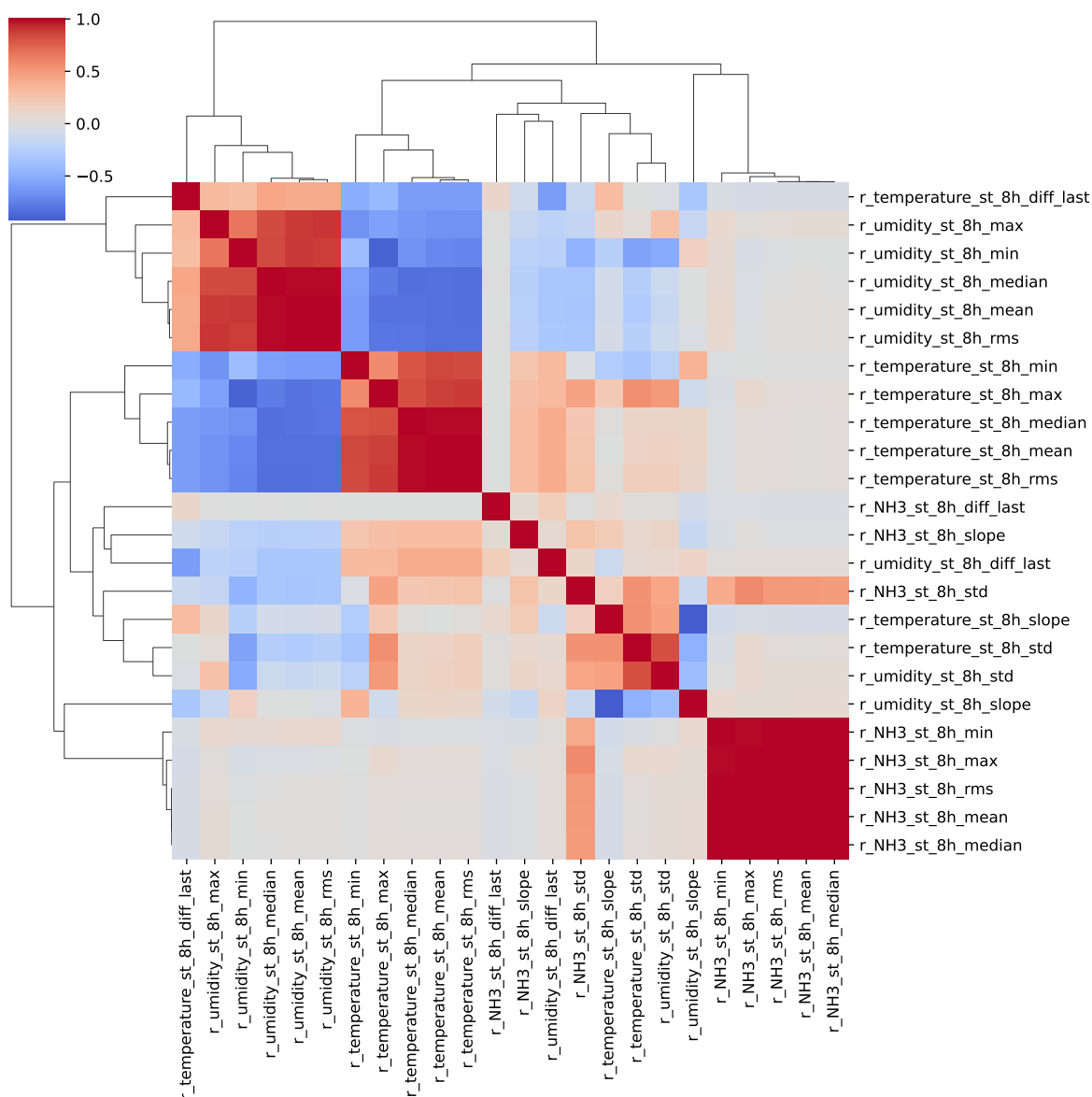
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_mean ; **cluster 7:** r_NH3_st_8h_std ; **cluster 8:** r_NH3_st_8h_diff_last ; **cluster 9:** r_NH3_st_8h_slope ; **cluster 10:** r_umidity_st_8h_mean, r_umidity_st_8h_median, r_temperature_st_8h_rms, r_temperature_st_8h_mean, r_umidity_st_8h_max, r_temperature_st_8h_max, r_temperature_st_8h_median, r_umidity_st_8h_min, r_temperature_st_8h_min, r_umidity_st_8h_rms ; **cluster 11:** r_umidity_st_8h_std, r_temperature_st_8h_std ; **cluster 12:** r_temperature_st_8h_diff_last ; **cluster 13:** r_temperature_st_8h_slope, r_umidity_st_8h_slope ; **cluster 14:** r_umidity_st_8h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_rms, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_rms, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_mean,

r_umidity_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope,
r_umidity_st_8h_diff_last

Len(Features_X): 14

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model':
None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, ran-
dom_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

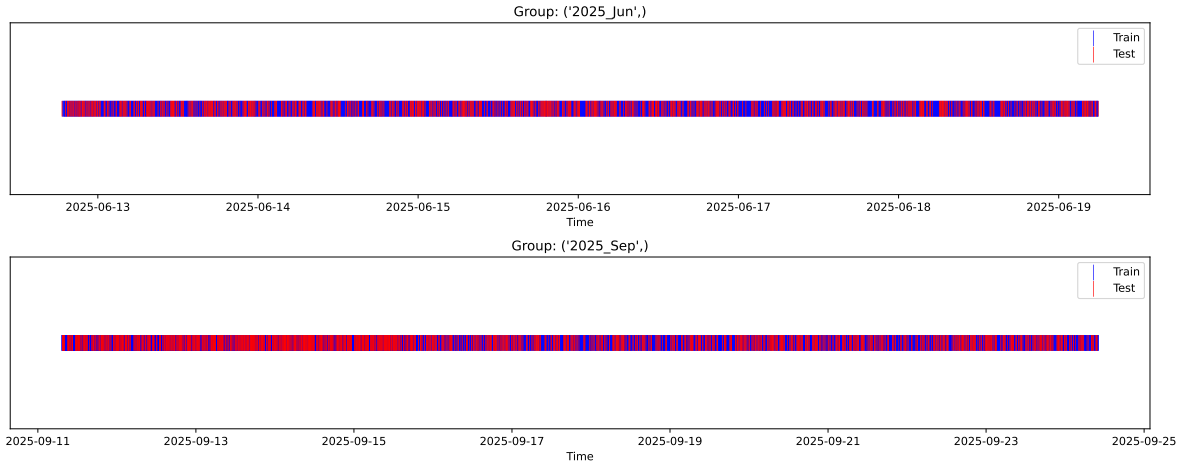
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5513 samples Test: 2363 samples



3.3.2 Internal CV Folds

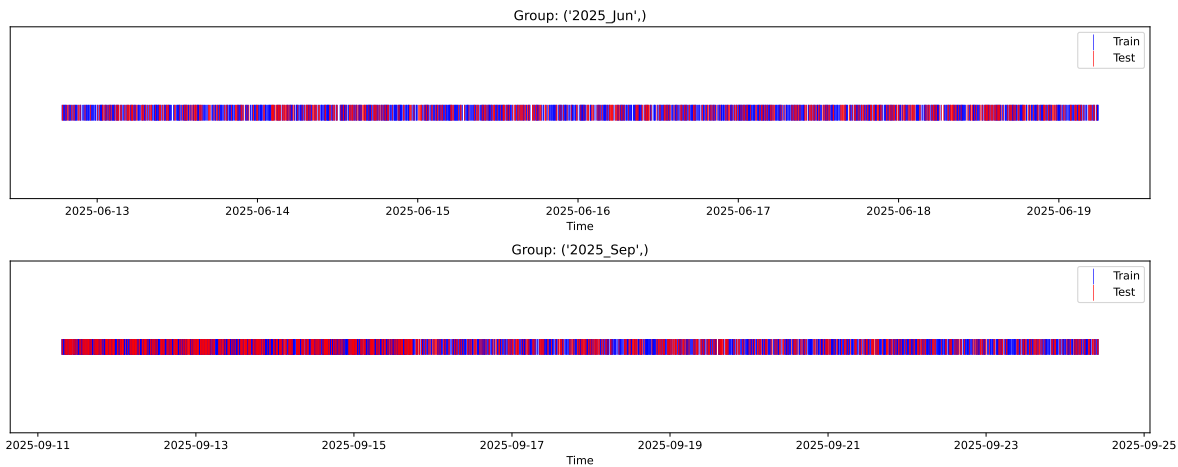
Fold 1/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.811, min=0.959, max=13.845
 y_{test} : mean=2.358, std=0.777, min=1.068, max=7.822

Fold 0 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.35, std=0.82

Fold 1 — y_{test} : mean=2.32, std=0.78, y_{train} : mean=2.37, std=0.82

Fold 2 — y_{test} : mean=2.35, std=0.85, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.36, std=0.78, y_train: mean=2.35, std=0.83

Fold 4 — y_test: mean=2.32, std=0.75, y_train: mean=2.37, std=0.84

train scores CV: [0.51306309 0.51716814 0.53152448 0.51428998 0.51327399]

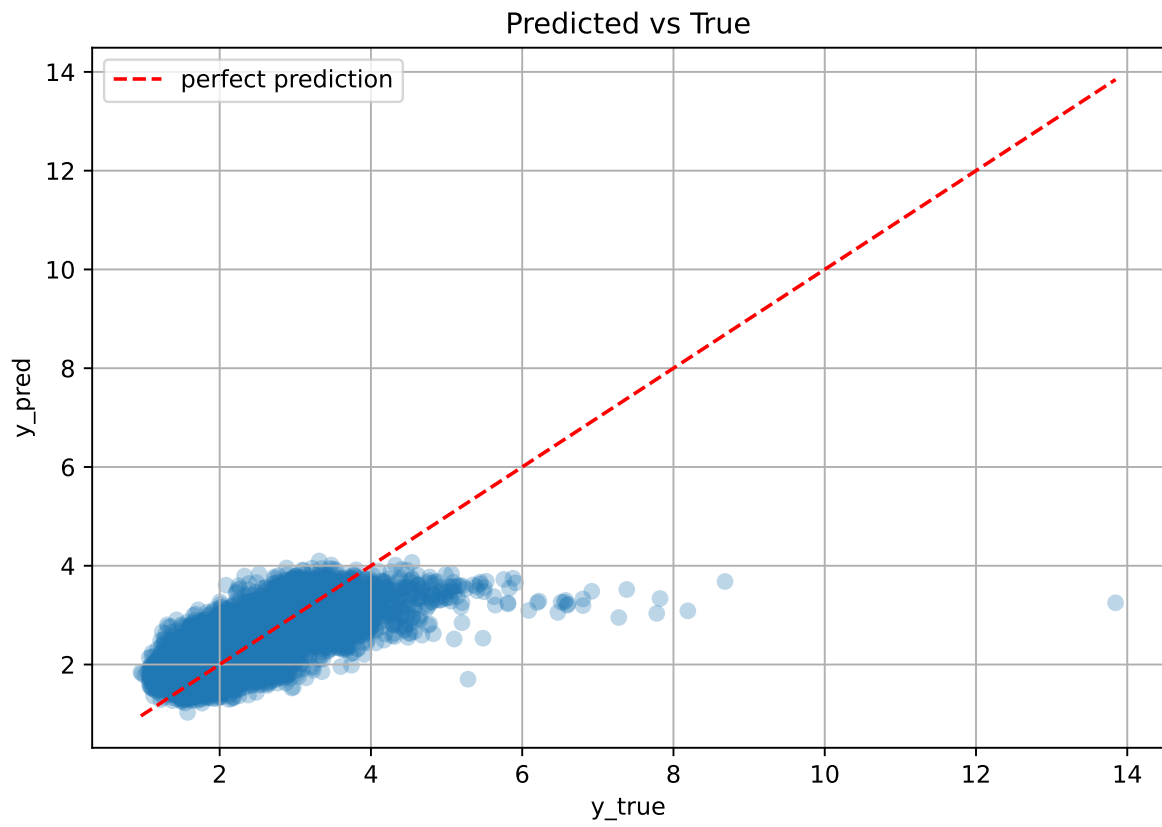
test scores CV: [0.52385075 0.51205242 0.48410699 0.52031162 0.52146544]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5179	0.5124	0.0147	0.5576	0.5588	0.0465	1.0907	1.0107

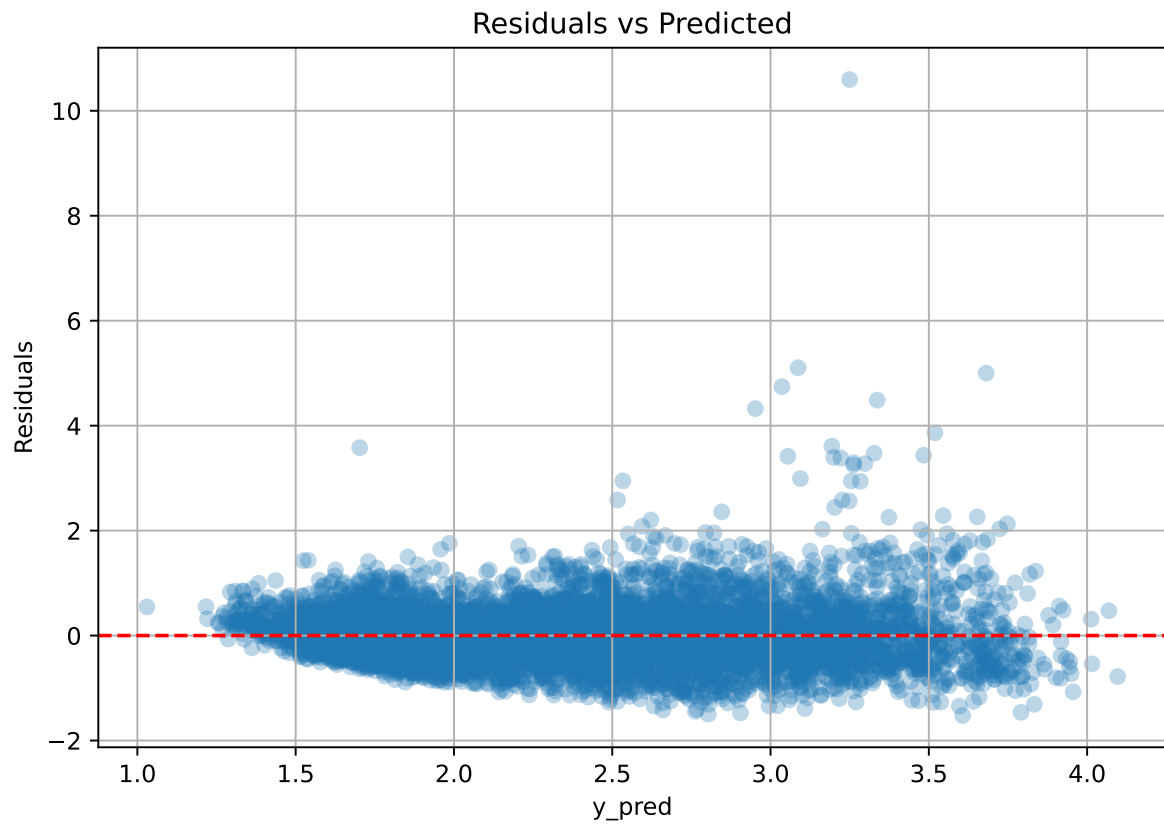
Coefficients:

coef_r_CO2 = -0.0512, coef_r_NH3 = -0.1849, coef_r_NH3_st_8h_diff_last = 0.0132, coef_r_NH3_st_8h_rms = 0.1051, coef_r_NH3_st_8h_slope = 0.0755, coef_r_NH3_st_8h_std = 0.0561, coef_r_THI = 0.5610, coef_r_temperature = -0.4459, coef_r_temperature_st_8h_diff_last = 0.1697, coef_r_temperature_st_8h_slope = -0.2608, coef_r_umidity = -0.8200, coef_r_umidity_st_8h_diff_last = -0.0089, coef_r_umidity_st_8h_mean = 0.3497, coef_r_umidity_st_8h_std = -0.0886, intercept = 2.3540,

3.4.2 Scatter



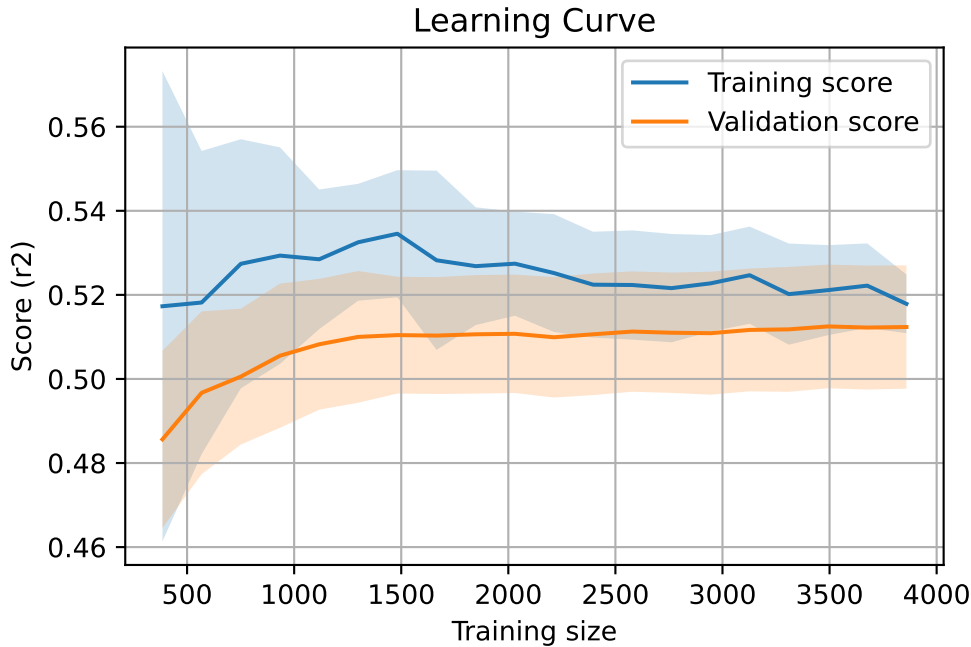
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362



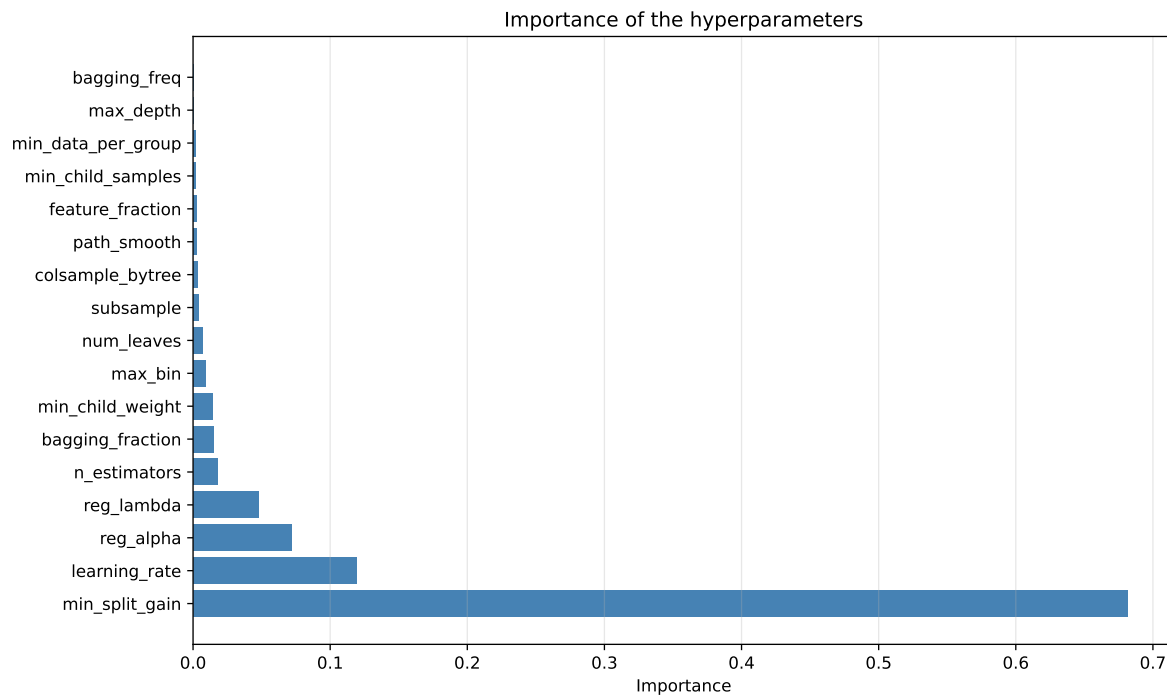
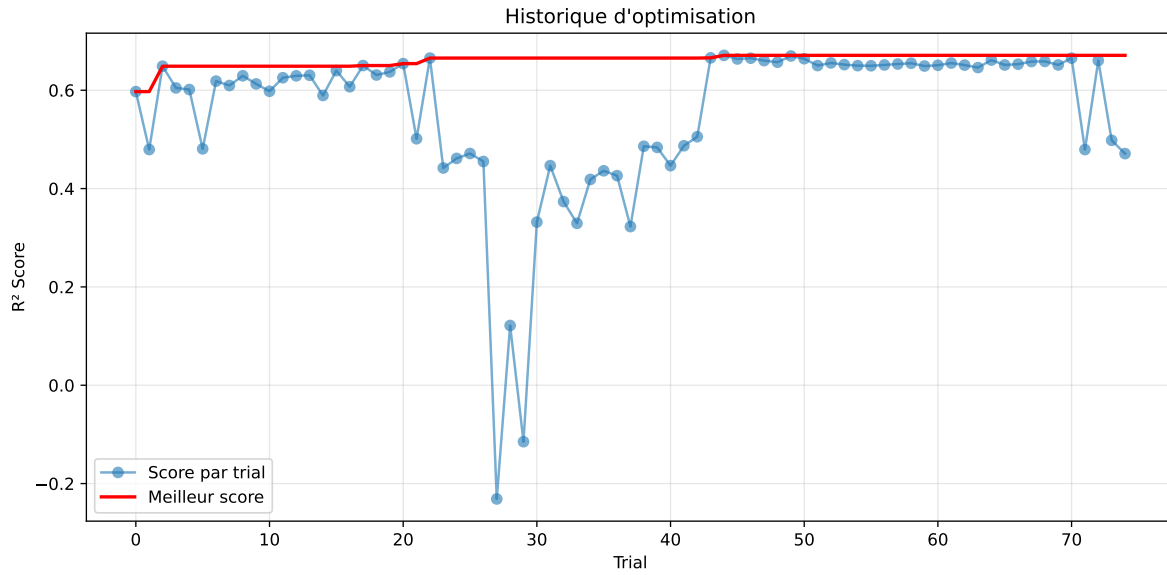
3.5 Model : LGBM

3.5.1 Gridsearch

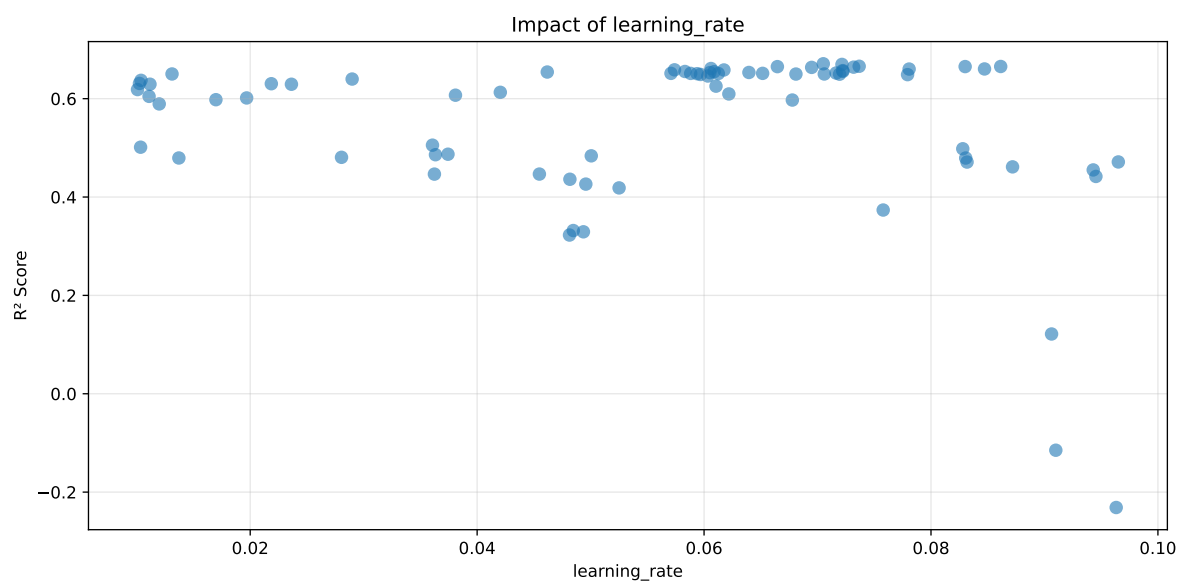
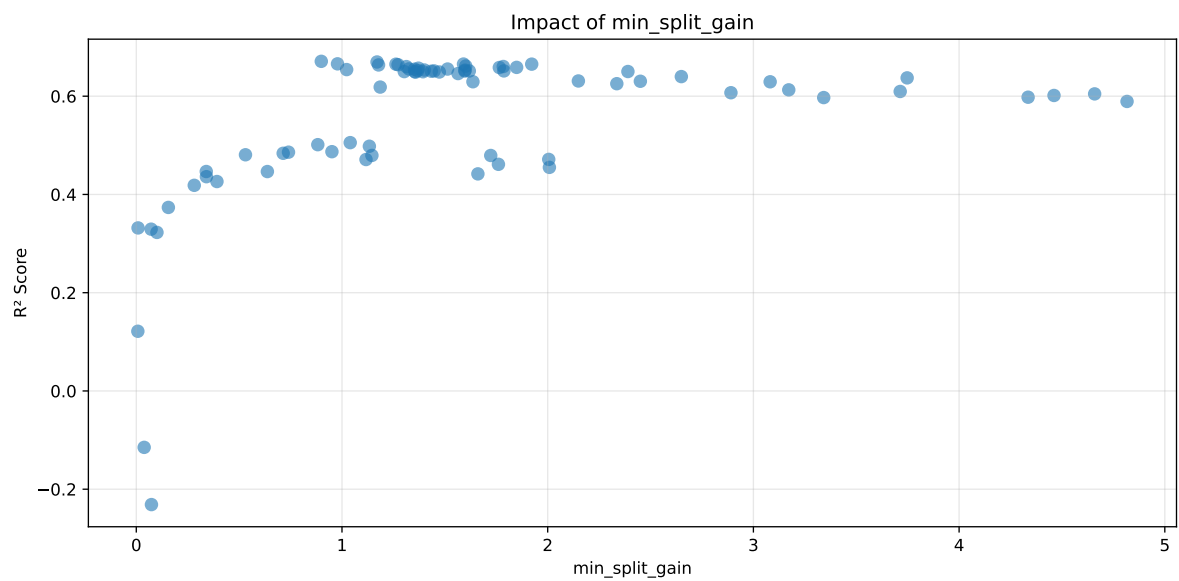
Distribution y_train vs y_test: y_train: mean=2.354, std=0.811, min=0.959, max=13.845
y_test: mean=2.358, std=0.777, min=1.068, max=7.822

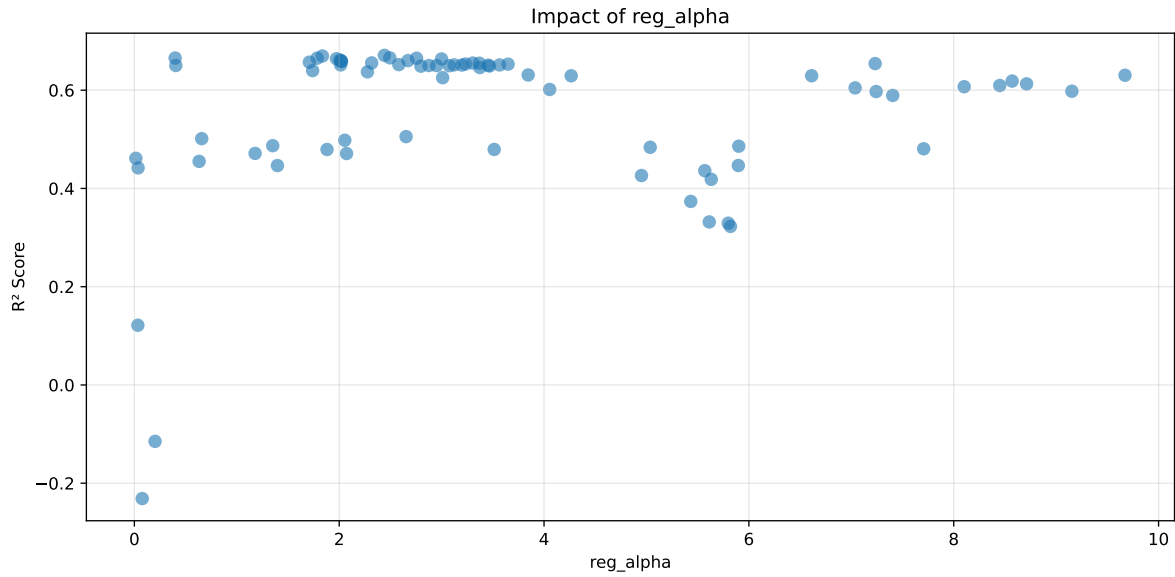
===== TOP Importance FEATURES =====

feature importance r_CO2 230 r_NH3_st_8h_rms 173 r_umidity 117 r_NH3 116 r_tem-
perature_st_8h_slope 98 r_umidity_st_8h_mean 80 r_temperature_st_8h_diff_last 79
r_temperature 72 r_NH3_st_8h_slope 56 r_NH3_st_8h_std 54 r_umidity_st_8h_diff_last
51 r_umidity_st_8h_std 46 r_NH3_st_8h_diff_last 45 r_THI 42



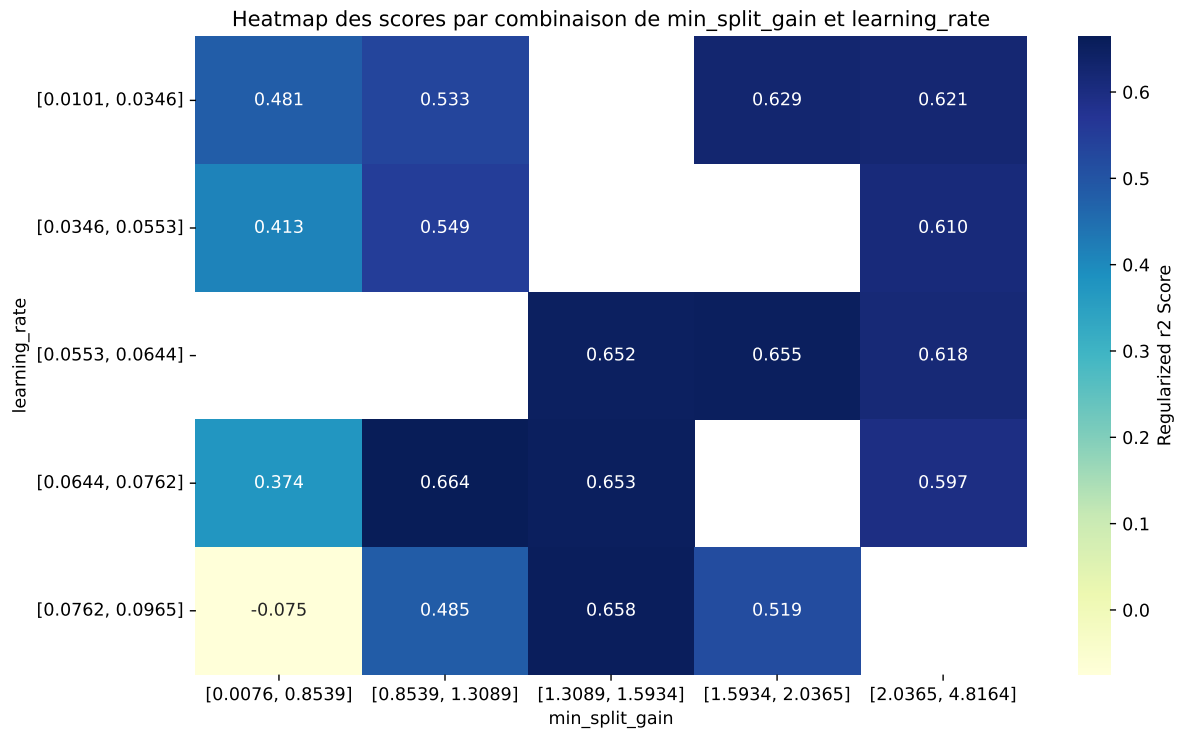
Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate', 'reg_alpha']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7013	0.671	0.6979	0.0208	0.671	0.7185	0.0476	1.0709	1.0452
2	0.703	0.6697	0.6961	0.0186	0.6697	0.7185	0.0489	1.073	1.0498
3	0.6967	0.6659	0.6934	0.0206	0.6659	0.7146	0.0487	1.0731	1.0464
4	0.698	0.6654	0.6918	0.0198	0.6654	0.7126	0.0472	1.0709	1.049
5	0.6953	0.6653	0.6886	0.0181	0.6653	0.7138	0.0486	1.073	1.0452
6	0.6959	0.6652	0.6927	0.0185	0.6652	0.7093	0.0441	1.0664	1.0462
7	0.6936	0.6641	0.6903	0.019	0.6641	0.713	0.0489	1.0737	1.0444
8	0.6939	0.6636	0.6913	0.0199	0.6636	0.7123	0.0487	1.0734	1.0457
9	0.6917	0.6613	0.6866	0.0181	0.6613	0.7096	0.0482	1.0729	1.0459
10	0.6908	0.6604	0.6883	0.0213	0.6604	0.7072	0.0467	1.0708	1.046
---	---	---	---	---	---	---	---	---	---
75	0.9522	0.755	0.775	0.0182	-0.2313	0.7991	0.0441	1.0584	1.2613

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.07050681653540908, 'reg__num_leaves': 61, 'reg__subsample': 0.5819310830889144, 'reg__colsample_bytree': 0.7810392388141231, 'reg__min_child_samples': 22, 'reg__reg_alpha': 2.4409533158647596, 'reg__reg_lambda': 46.438029892801126, 'reg__min_split_gain': 0.8993831771513598, 'reg__path_smooth': 3.2199312210282405, 'reg__min_child_weight': 0.0028139940639143658, 'reg__feature_fraction': 0.556553217064878, 'reg__bagging_fraction': 0.5435312594386981, 'reg__bagging_freq': 3, 'reg__max_bin': 160, 'reg__min_data_per_group': 136}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.072154188325569, 'reg__num_leaves': 38, 'reg__subsample': 0.5771878996435069, 'reg__colsample_bytree': 0.8042559734850646, 'reg__min_child_samples': 10, 'reg__reg_alpha': 1.8358241674153606, 'reg__reg_lambda': 45.03494669639166, 'reg__min_split_gain': 1.1704265807403849, 'reg__path_smooth': 2.074986212412472, 'reg__min_child_weight': 0.02515766031508471, 'reg__feature_fraction': 0.5414095768449059, 'reg__bagging_fraction': 0.642322390437486, 'reg__bagging_freq': 3, 'reg__max_bin': 168, 'reg__min_data_per_group': 135}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.07367049426129124, 'reg__num_leaves': 40, 'reg__subsample': 0.574813019953608, 'reg__colsample_bytree': 0.7899397165051106, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.4942522462885237, 'reg__reg_lambda': 47.47701090797765, 'reg__min_split_gain': 0.9782503035376563, 'reg__path_smooth': 2.6348511784408517, 'reg__min_child_weight': 0.02441286457186155, 'reg__feature_fraction': 0.5473753776012219, 'reg__bagging_fraction': 0.5461756327893039, 'reg__bagging_freq': 3, 'reg__max_bin': 159, 'reg__min_data_per_group': 135}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.0861426871525584, 'reg__num_leaves': 52, 'reg__subsample': 0.7728869201916477, 'reg__colsample_bytree': 0.672634101741693, 'reg__min_child_samples': 10, 'reg__reg_alpha': 0.3981219794041424, 'reg__reg_lambda': 44.057348018931584, 'reg__min_split_gain': 1.590696995489785, 'reg__path_smooth': 3.6013482286559526, 'reg__min_child_weight': 0.0018702676071806598, 'reg__feature_fraction': 0.5041667310621232, 'reg__bagging_fraction': 0.7158036153763856, 'reg__bagging_freq': 3, 'reg__max_bin': 192, 'reg__min_data_per_group': 144}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.08300989923584245, 'reg__num_leaves': 59, 'reg__subsample': 0.5010603176727811, 'reg__colsample_bytree': 0.7809728105635855, 'reg__min_child_samples': 33, 'reg__reg_alpha': 1.7850600113754123, 'reg__reg_lambda': 21.636801999402355, 'reg__min_split_gain': 1.9222950837802135, 'reg__path_smooth': 1.2251005739838345, 'reg__min_child_weight': 0.0052622152722669804, 'reg__feature_fraction': 0.5897020026016595, 'reg__bagging_fraction': 0.7658962771957654, 'reg__bagging_freq': 2, 'reg__max_bin': 139, 'reg__min_data_per_group': 123}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.06646215146058922, 'reg__num_leaves': 38, 'reg__subsample': 0.5631310601803248, 'reg__colsample_bytree': 0.7921875782653162, 'reg__min_child_samples': 15, 'reg__reg_alpha': 2.7553781375778295, 'reg__reg_lambda': 27.954699438073032, 'reg__min_split_gain': 1.2622171870992132, 'reg__path_smooth': 4.638104492729499, 'reg__min_child_weight': 0.016099948176860564, 'reg__feature_fraction': 0.5476809309852748, 'reg__bagging_fraction': 0.5540313496477046, 'reg__bagging_freq': 2, 'reg__max_bin': 167, 'reg__min_data_per_group': 130}

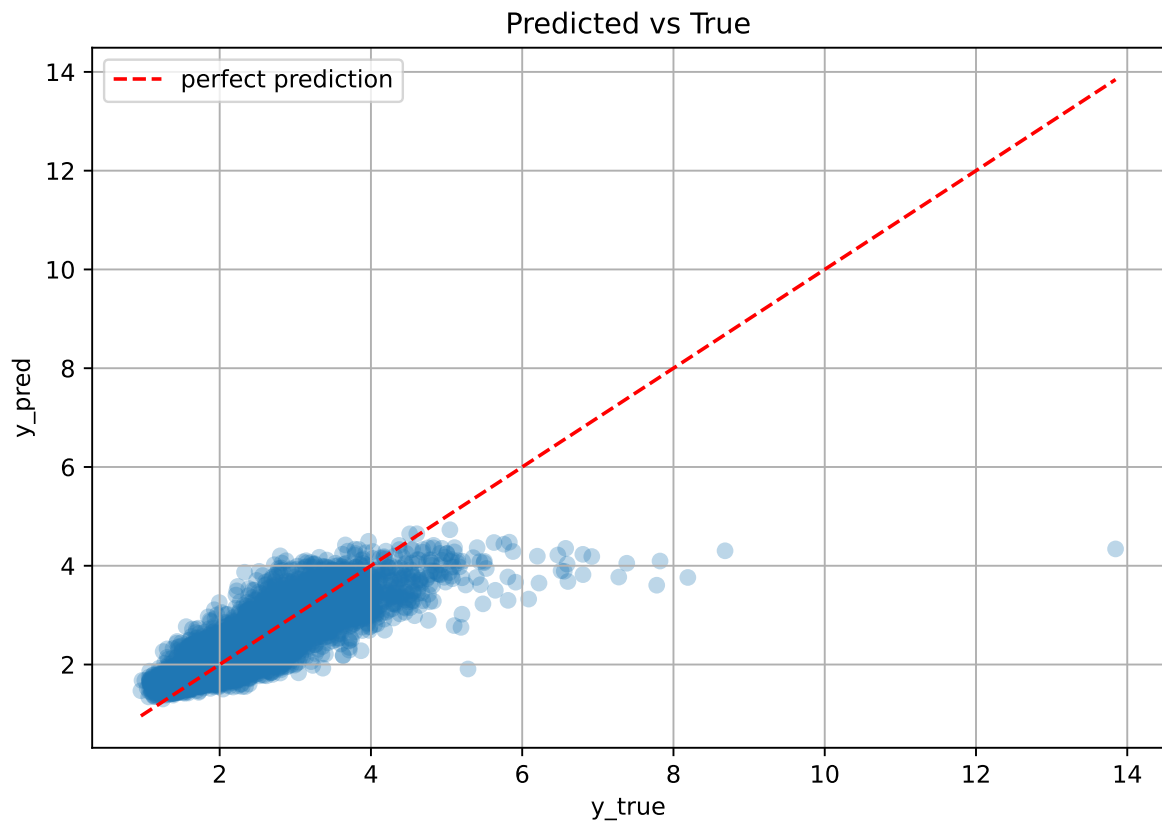
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.0731816860897162, 'reg__num_leaves': 63, 'reg__subsample': 0.5794545561993908, 'reg__colsample_bytree': 0.795245643625183, 'reg__min_child_samples': 25, 'reg__reg_alpha': 1.9733931514990053, 'reg__reg_lambda': 45.06421985744912, 'reg__min_split_gain': 1.2726130557717608, 'reg__path_smooth': 1.9366773776499426, 'reg__min_child_weight': 0.002111827312239421, 'reg__feature_fraction': 0.5339775351271544, 'reg__bagging_fraction': 0.6540444424621323, 'reg__bagging_freq': 3, 'reg__max_bin': 154, 'reg__min_data_per_group': 186}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.0694823371378364, 'reg__num_leaves': 39, 'reg__subsample': 0.5739071255460388, 'reg__colsample_bytree': 0.7883750537082692, 'reg__min_child_samples': 22, 'reg__reg_alpha': 2.9994306094990915, 'reg__reg_lambda': 45.53868309648463, 'reg__min_split_gain': 1.1777340884741587, 'reg__path_smooth': 2.612031205811662, 'reg__min_child_weight': 0.002751433958586304, 'reg__feature_fraction': 0.5598455828782072, 'reg__bagging_fraction': 0.65268082385751, 'reg__bagging_freq': 2, 'reg__max_bin': 163, 'reg__min_data_per_group': 133}

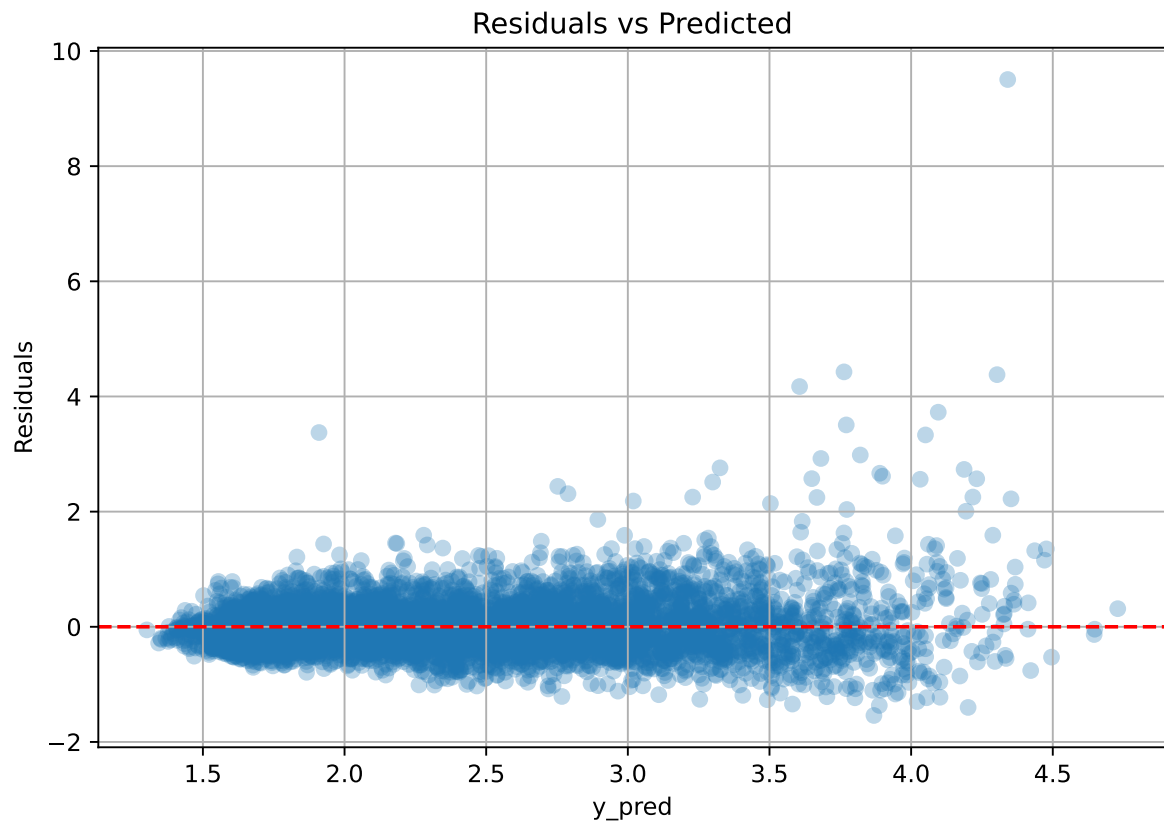
Rank 9: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.06060998549487476, 'reg__num_leaves': 32, 'reg__subsample': 0.5359010585514644, 'reg__colsample_bytree': 0.8481769371729995, 'reg__min_child_samples': 12, 'reg__reg_alpha': 2.0066374039065407, 'reg__reg_lambda': 21.886087780431335, 'reg__min_split_gain': 1.6015533006694473, 'reg__path_smooth': 1.247285547882956, 'reg__min_child_weight': 0.006304163893259626, 'reg__feature_fraction': 0.5201447193762042, 'reg__bagging_fraction': 0.615558478323301, 'reg__bagging_freq': 3, 'reg__max_bin': 135, 'reg__min_data_per_group': 190}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.0847171957090203, 'reg__num_leaves': 59, 'reg__subsample': 0.6013858180589795, 'reg__colsample_bytree': 0.777426256381563, 'reg__min_child_samples': 33, 'reg__reg_alpha': 2.018292960126684, 'reg__reg_lambda': 36.2941631521858, 'reg__min_split_gain': 1.7841143311895982, 'reg__path_smooth': 2.472413575084005, 'reg__min_child_weight': 0.0019320801019174303, 'reg__feature_fraction': 0.5941229260050876, 'reg__bagging_fraction': 0.7740855100763001, 'reg__bagging_freq': 2, 'reg__max_bin': 172, 'reg__min_data_per_group': 123}

3.5.2 Scatter



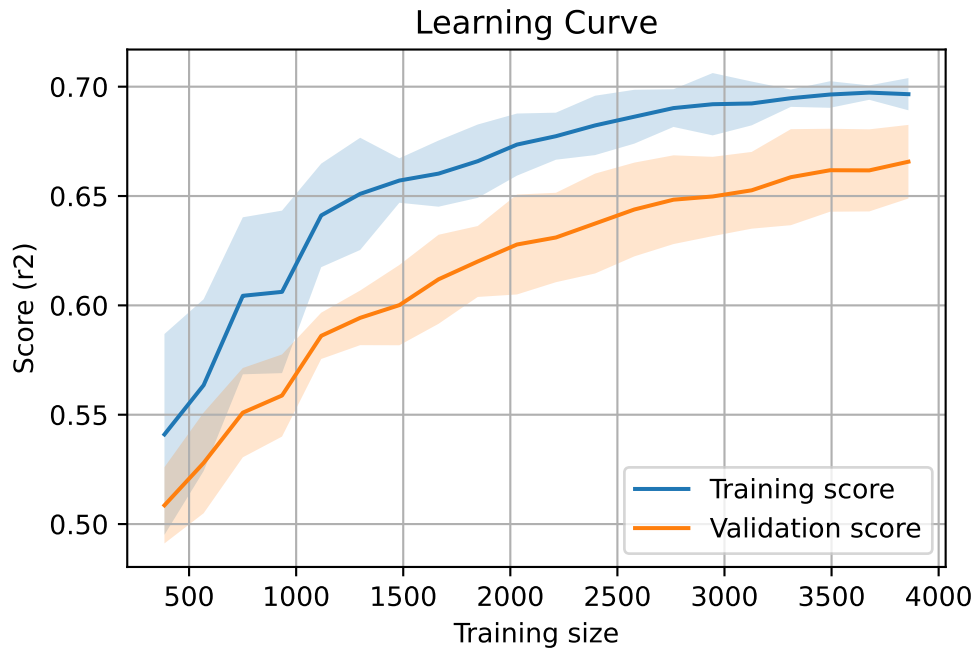
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7876, **Len(training)** : 5514,

Len(training_real) : 3860, **Len(test_of_training)** : 1654, **Len(test)** : 2362

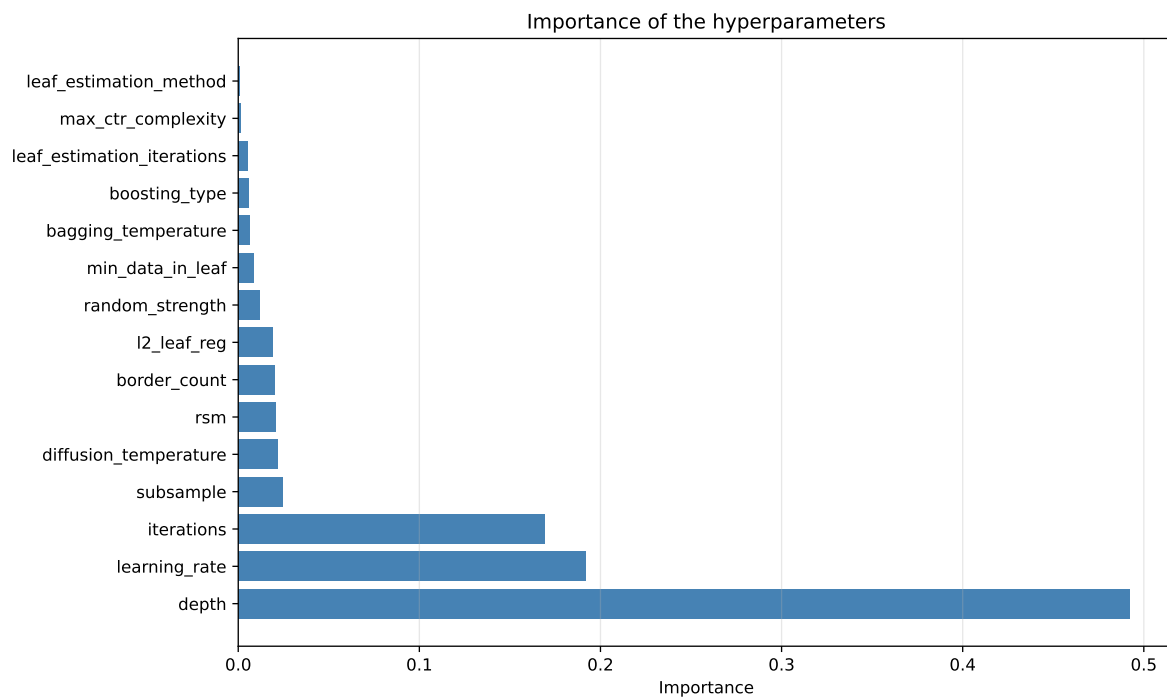
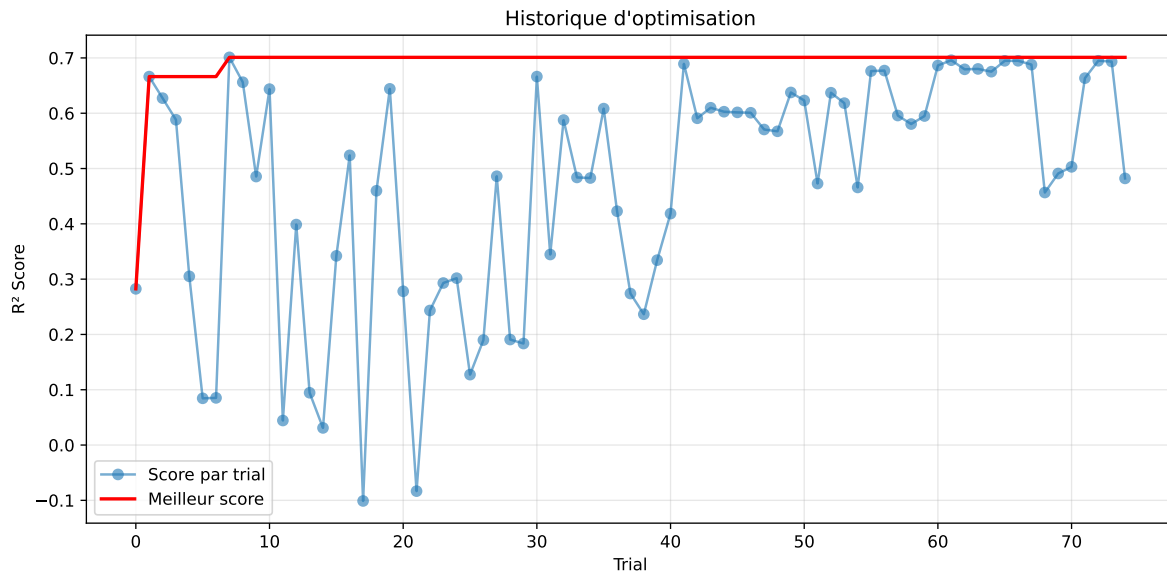


3.6 Model : CatBoost

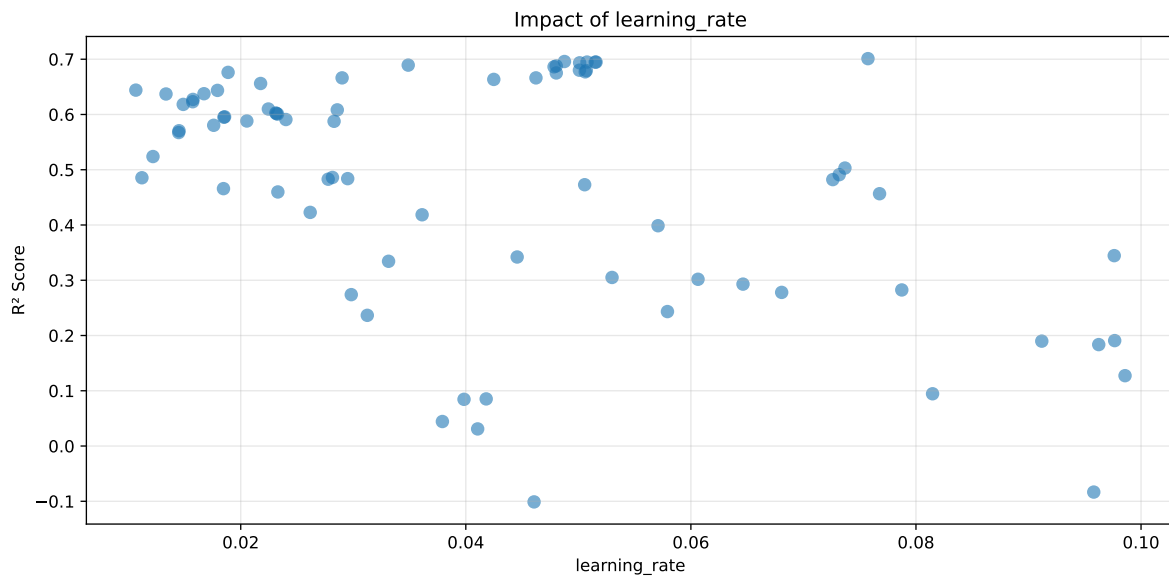
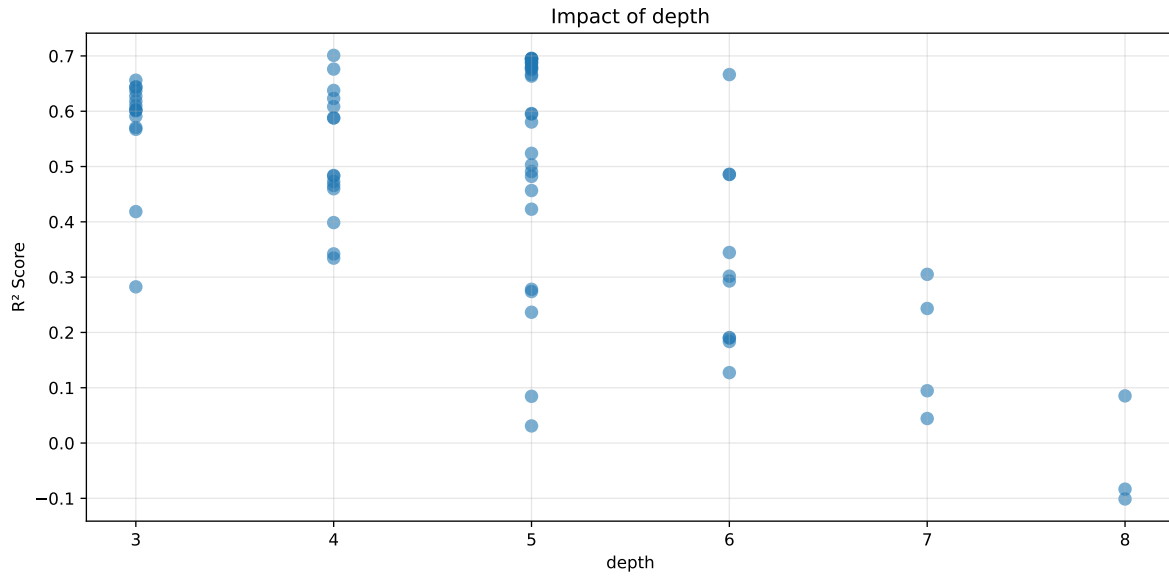
3.6.1 Gridsearch

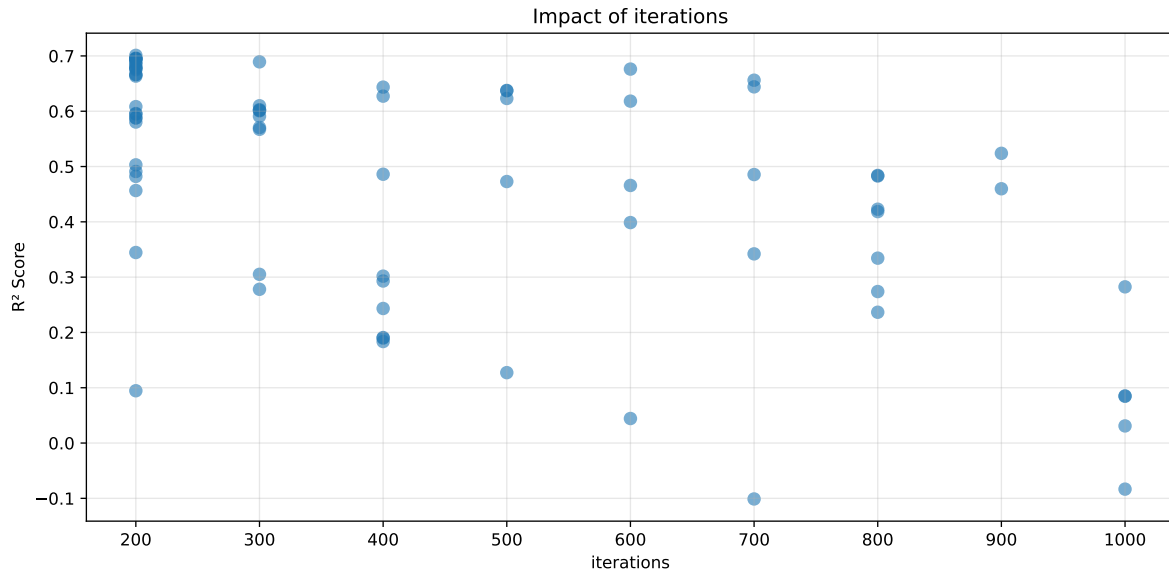
Distribution y_train vs y_test: y_train: mean=2.354, std=0.811, min=0.959, max=13.845
y_test: mean=2.358, std=0.777, min=1.068, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 22.523800 r_temperature_st_8h_diff_last 18.527235 r_CO2
9.381535 r_NH3 8.344617 r_NH3_st_8h_rms 6.957726 r_temperature_st_8h_slope
6.499262 r_umidity_st_8h_mean 5.657075 r_temperature 5.517421 r_NH3_st_8h_slope
4.198742 r_THI 4.039944 r_umidity_st_8h_diff_last 3.142528 r_NH3_st_8h_std 2.827680
r_umidity_st_8h_std 1.239126 r_NH3_st_8h_diff_last 1.143308



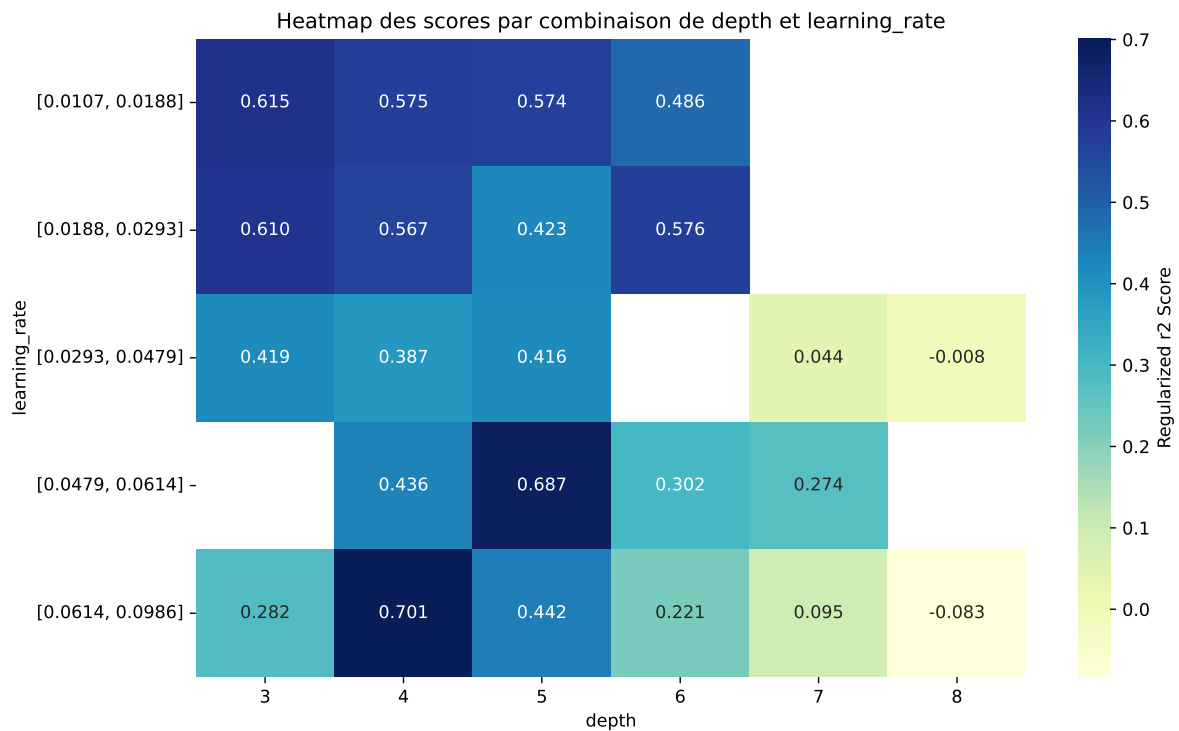
Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7349	0.701	0.7176	0.0173	0.701	0.7193	0.0182	1.026	1.0483
2	0.725	0.6959	0.714	0.0178	0.6959	0.7178	0.022	1.0316	1.0418
3	0.7255	0.6949	0.7143	0.0204	0.6949	0.7194	0.0246	1.0353	1.044
4	0.726	0.6948	0.7154	0.0215	0.6948	0.719	0.0242	1.0348	1.0449
5	0.7226	0.6948	0.7127	0.0174	0.6948	0.7186	0.0238	1.0343	1.04
6	0.7241	0.6935	0.713	0.0189	0.6935	0.7186	0.0251	1.0361	1.0441
7	0.7154	0.6893	0.7091	0.0199	0.6893	0.7124	0.0231	1.0335	1.0378
8	0.7149	0.6879	0.7082	0.0184	0.6879	0.7141	0.0262	1.0381	1.0392
9	0.7126	0.6863	0.7093	0.0192	0.6863	0.7104	0.0241	1.0351	1.0383
10	0.7034	0.6801	0.7026	0.017	0.6801	0.7032	0.0231	1.0339	1.0343
---	---	---	---	---	---	---	---	---	---
75	0.9659	0.7881	0.7993	0.024	-0.1011	0.8169	0.0288	1.0365	1.2257

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.07573492502850426, 'reg__l2_leaf_reg': 37.613704243863985, 'reg__bagging_temperature': 0.05227659035359211, 'reg__random_strength': 2.498317844989298, 'reg__subsample': 0.7162708062469858, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5170889947494868, 'reg__border_count': 104, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3503.638486076648}

Rank 2: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.048765992793673475, 'reg__l2_leaf_reg': 21.9194154374136, 'reg__bagging_temperature': 3.0700201550298494, 'reg__random_strength': 1.9666668490275763, 'reg__subsample': 0.7212482508556068, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6481736950998316, 'reg__border_count': 95, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6678.36268185777}

Rank 3: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.05077075018082384, 'reg__l2_leaf_reg': 38.48854212840165, 'reg__bagging_temperature': 3.0805418580608315, 'reg__random_strength': 1.964043040219483, 'reg__subsample': 0.7310275478719563, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6448697080263736, 'reg__border_count': 95, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4843.797613662298}

Rank 4: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.051575027578057084, 'reg__l2_leaf_reg': 39.65562486240354, 'reg__bagging_temperature': 2.533881325940205, 'reg__random_strength': 1.9063048078122211, 'reg__subsample': 0.7422268761718843, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 5, 'reg__rsm':

0.648459453230468, 'reg__border_count': 95, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4809.207872282291}

Rank 5: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.05149887189592918, 'reg__l2_leaf_reg': 39.58105528764312, 'reg__bagging_temperature': 2.8368538164637087, 'reg__random_strength': 1.944623596228164, 'reg__subsample': 0.7181152189861272, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.6500331210189573, 'reg__border_count': 67, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4991.244638003958}

Rank 6: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.05011464542118307, 'reg__l2_leaf_reg': 37.77594610652183, 'reg__bagging_temperature': 2.7762681523577584, 'reg__random_strength': 2.0193683987408977, 'reg__subsample': 0.7195441208486286, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6442066215329963, 'reg__border_count': 95, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7794.125824576746}

Rank 7: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.034882868679920166, 'reg__l2_leaf_reg': 42.724468718291135, 'reg__bagging_temperature': 0.6842071769535174, 'reg__random_strength': 2.886791410882665, 'reg__subsample': 0.5057380161961763, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5635299368897844, 'reg__border_count': 90, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1767.5625331315441}

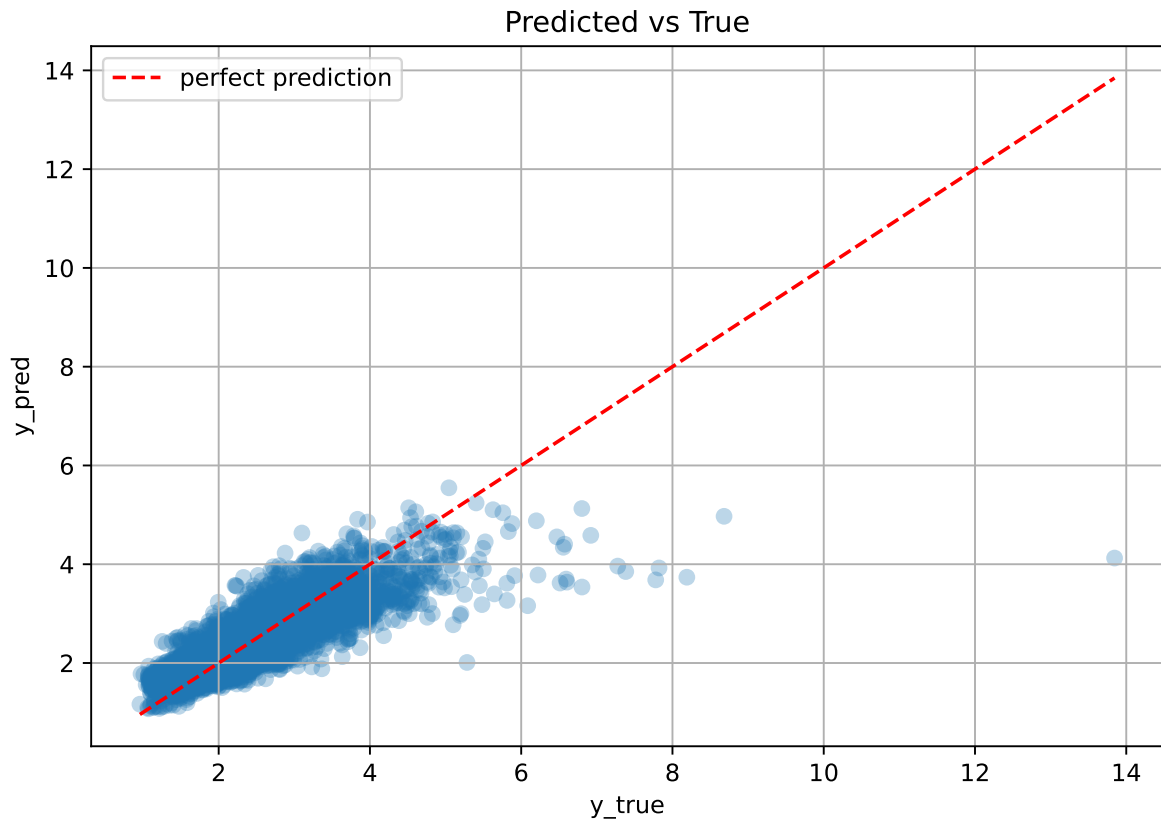
Rank 8: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.04803439250057682, 'reg__l2_leaf_reg': 38.83401690603761, 'reg__bagging_temperature': 2.59170666469534, 'reg__random_strength': 2.0079401664494614, 'reg__subsample': 0.7275691604574354, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.5457419250156129, 'reg__border_count': 71, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4839.4465096148715}

Rank 9: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.04784074735408661, 'reg__l2_leaf_reg': 38.27002361782286, 'reg__bagging_temperature': 1.671354391324577, 'reg__random_strength': 2.013261362507812, 'reg__subsample': 0.5298521511668987, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.7408604663454125, 'reg__border_count': 95, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 419.4900986561179}

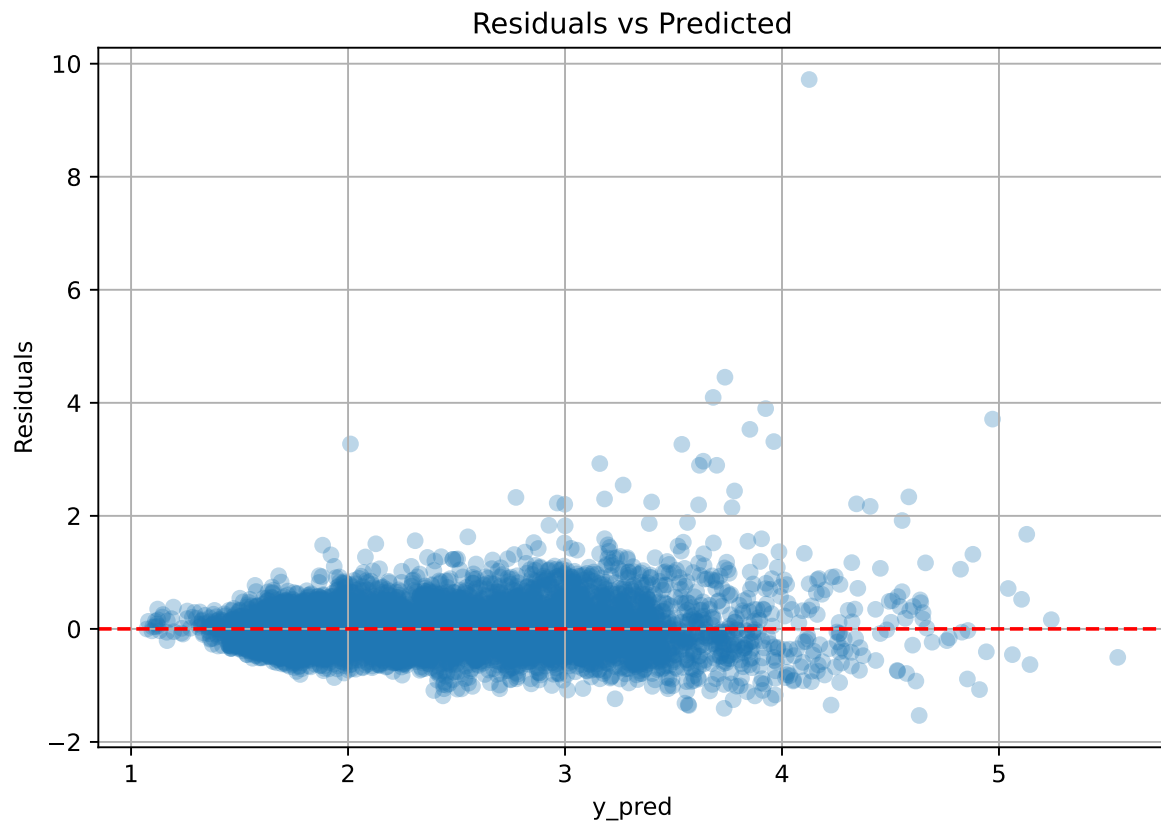
Rank 10: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.05009502036204562, 'reg__l2_leaf_reg': 47.40121602730506, 'reg__bagging_temperature': 2.9356652048405865,

'reg__random_strength': 4.1143727624538595, 'reg__subsample': 0.7234135813663573,
'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 7, 'reg__rsm':
0.6526225193565693, 'reg__border_count': 67, 'reg__leaf_estimation_method': 'Gradient',
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_tempera-
ture': 4782.935285962173}

3.6.2 Scatter



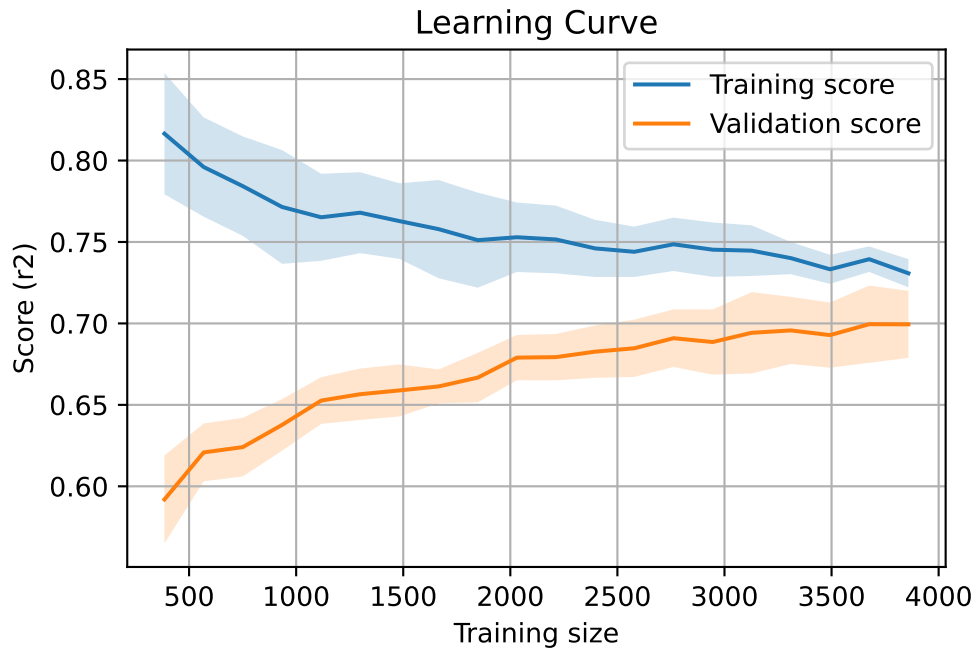
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.518	0.512	0.015	0.558	0.559	0.046	1.091	1.011	nan
LGBM	0.701	0.671	0.021	0.698	0.718	0.048	1.071	1.045	0.671
CatBoost	0.735	0.701	0.017	0.718	0.719	0.018	1.026	1.048	0.701

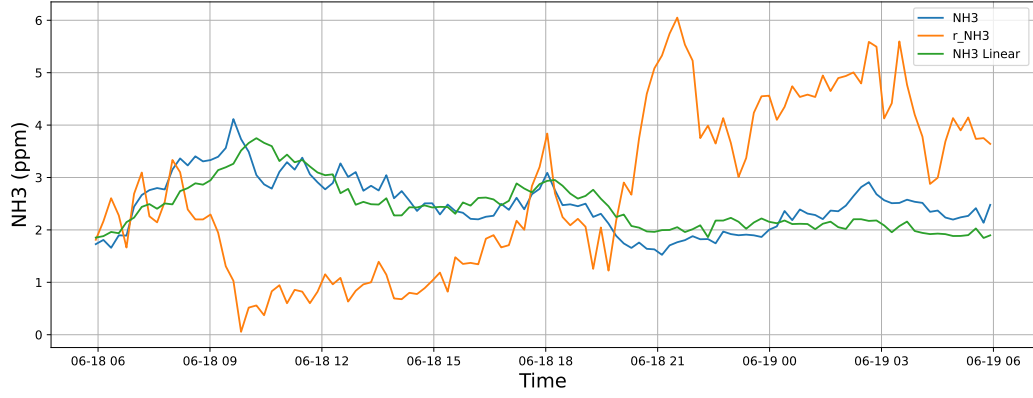
4 Plot

4.1 Standalone plots

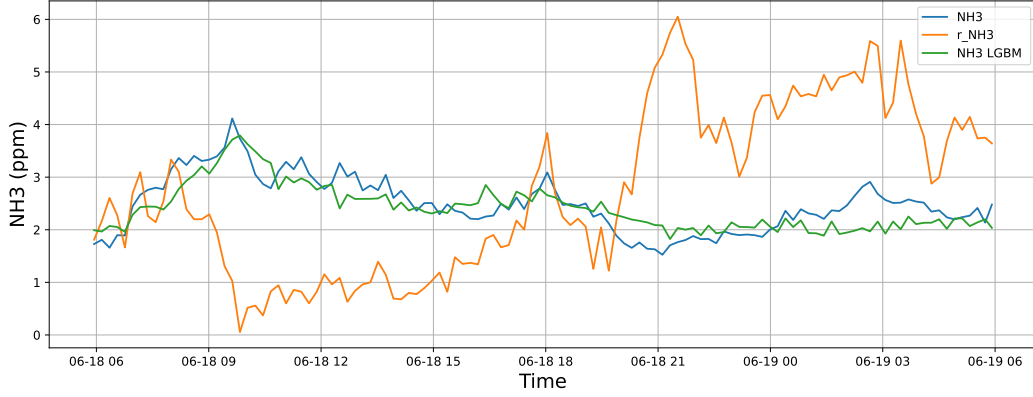
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

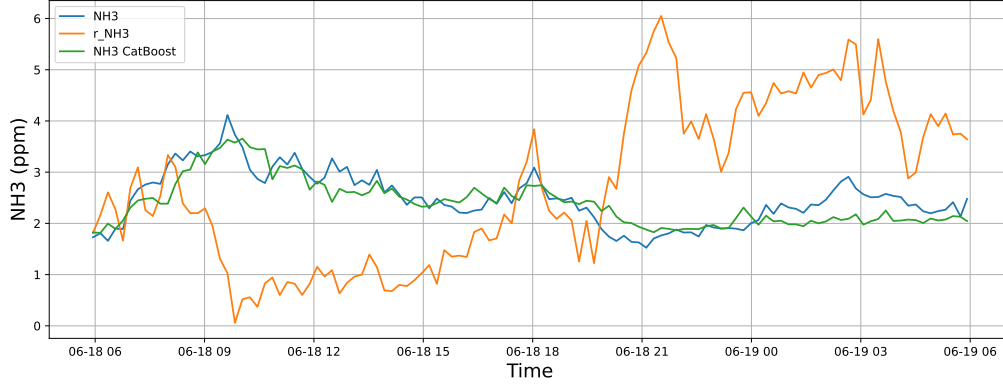
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

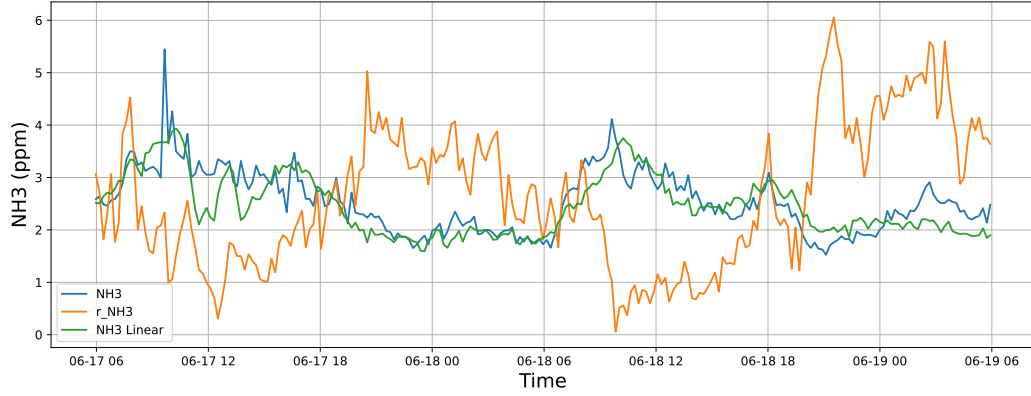


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

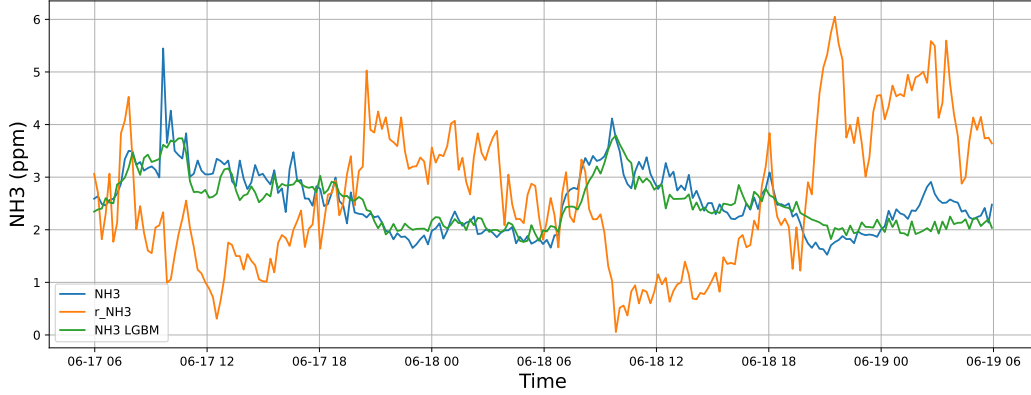


4.1.1.2 Time window : 48

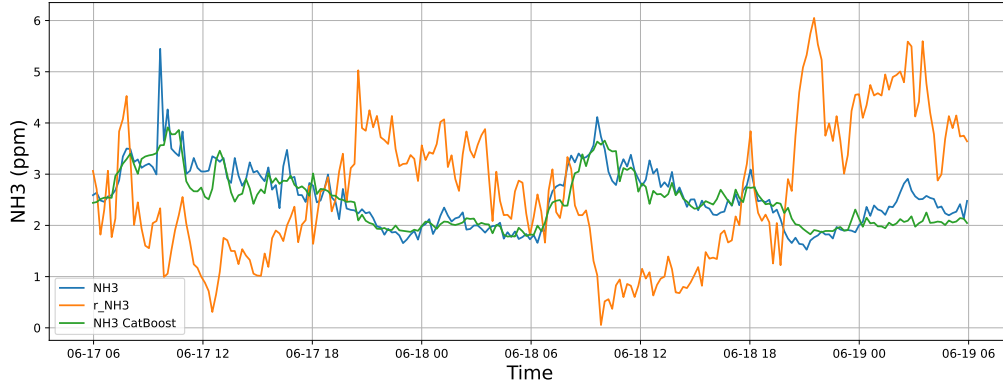
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

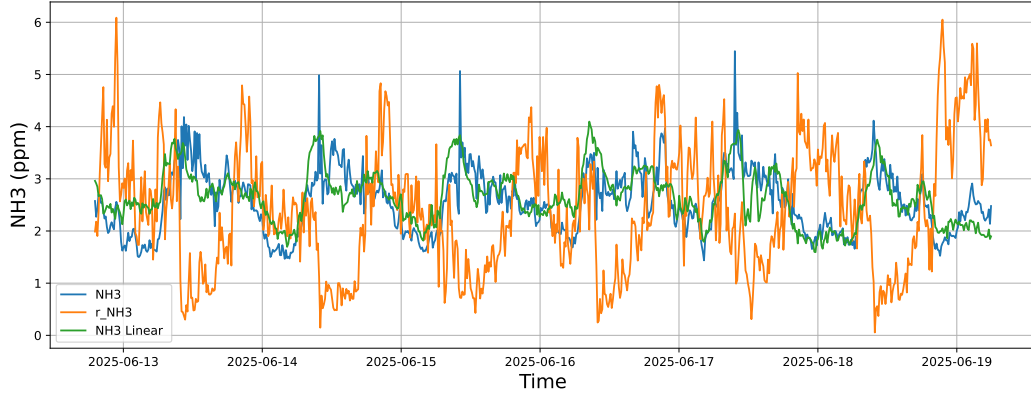


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

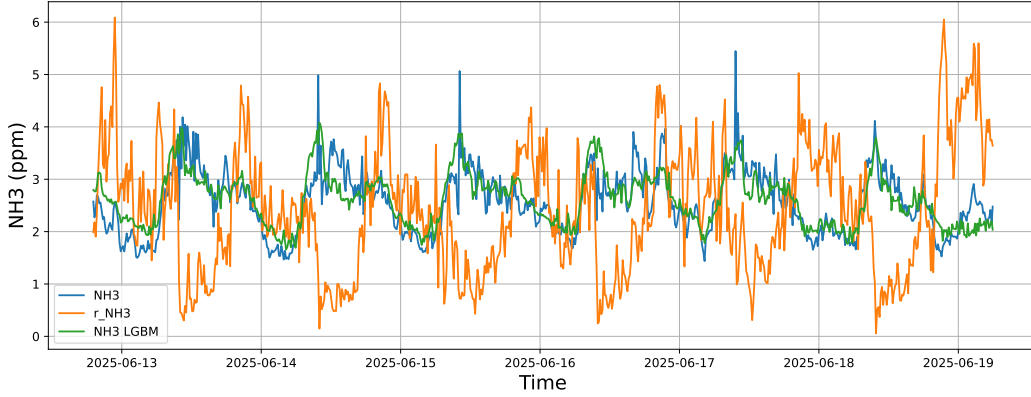


4.1.1.3 Time window : ALL

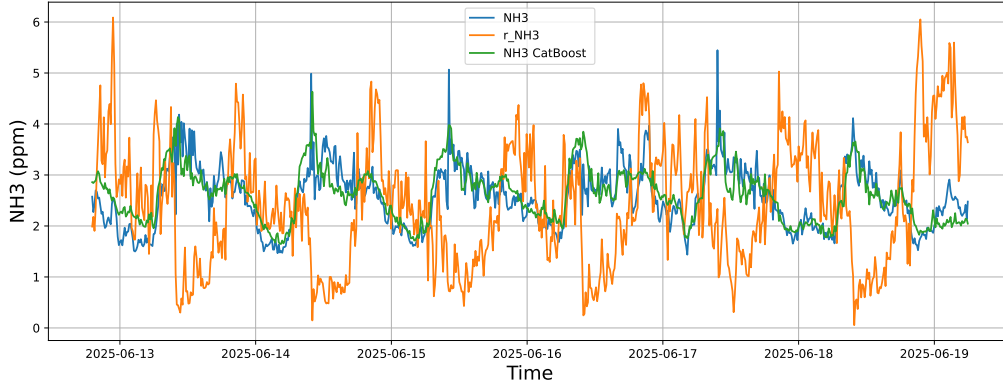
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



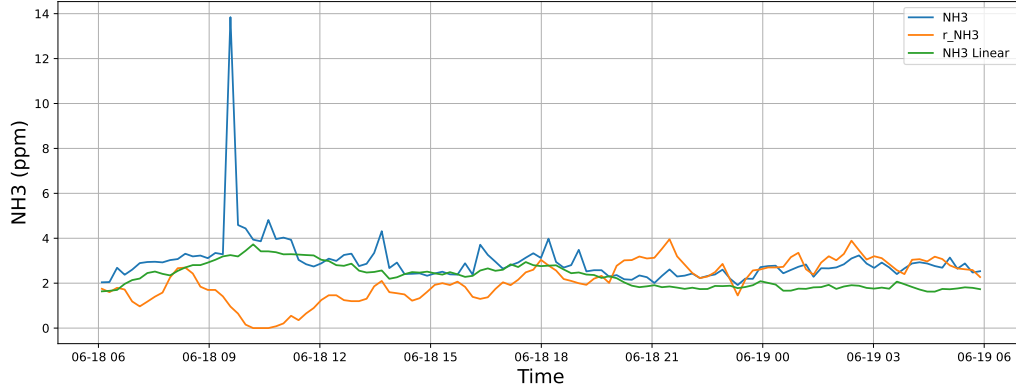
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



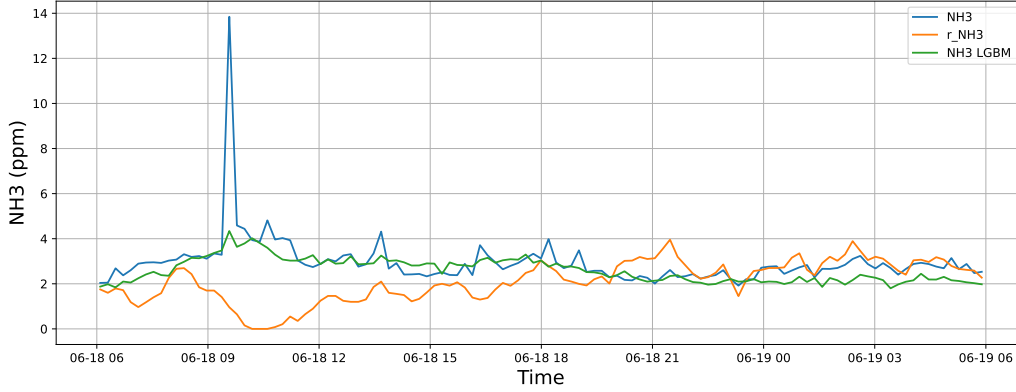
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

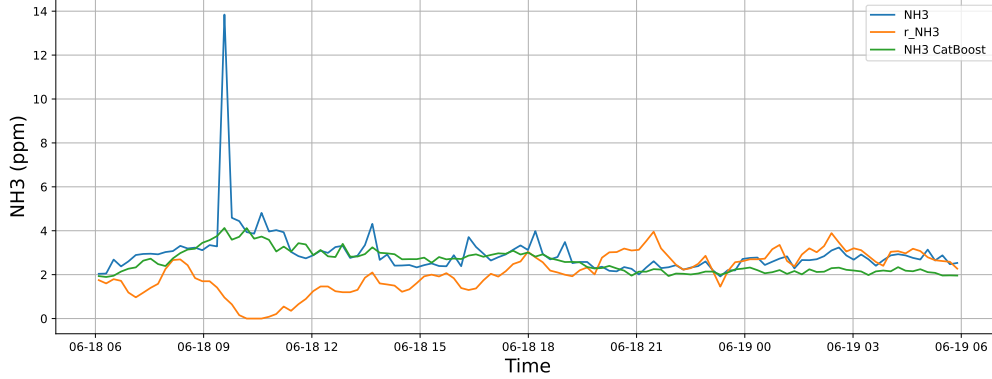
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

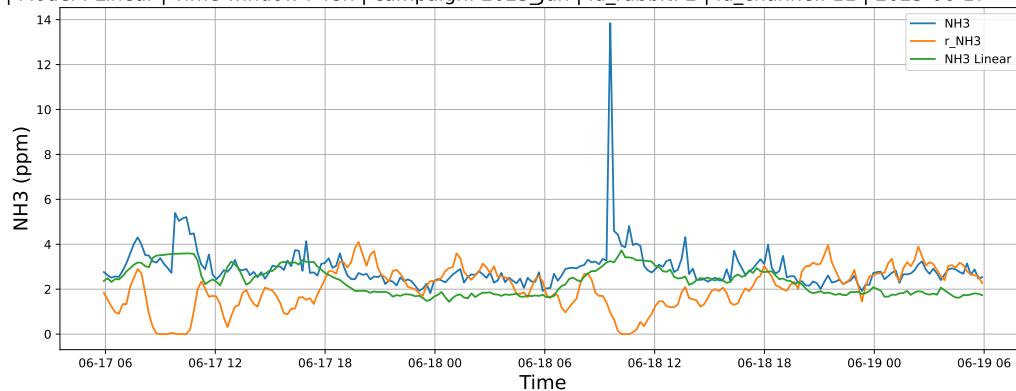


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

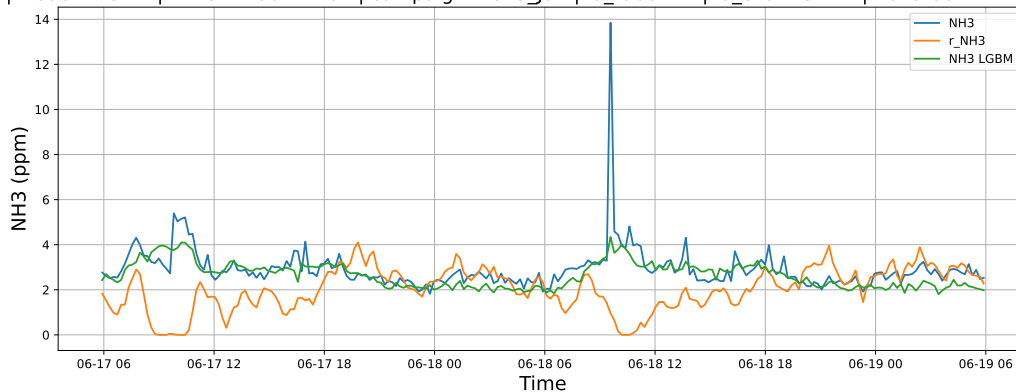


4.1.2.2 Time window : 48

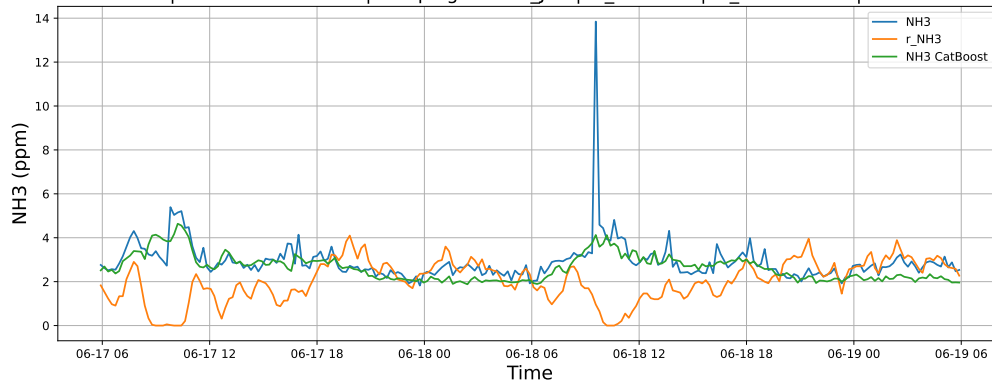
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

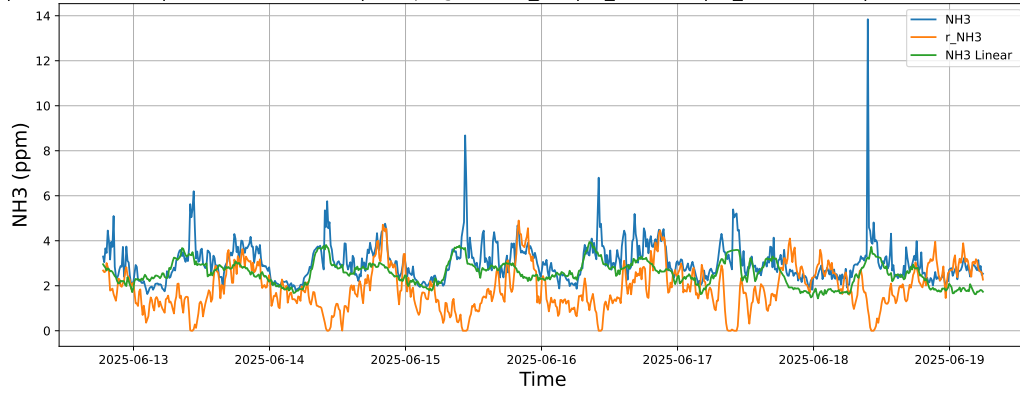


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

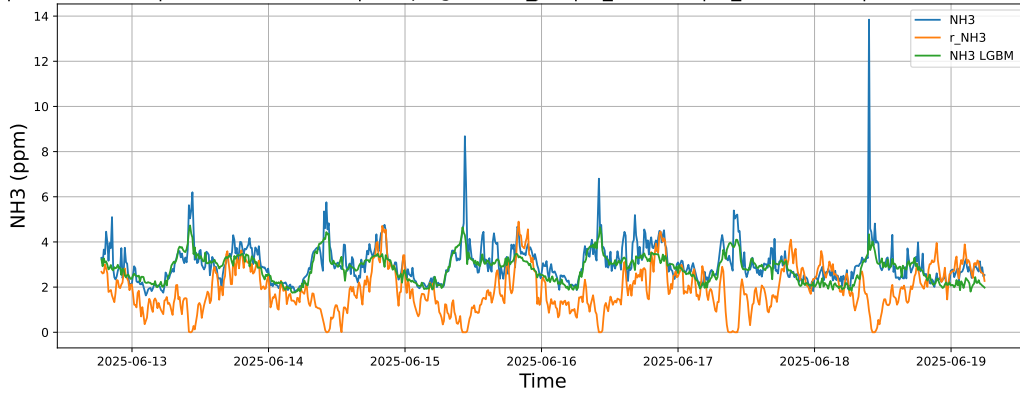


4.1.2.3 Time window : ALL

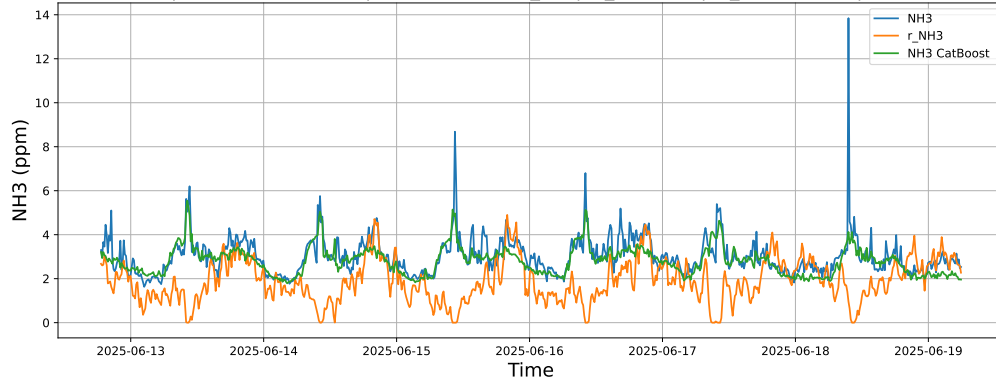
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



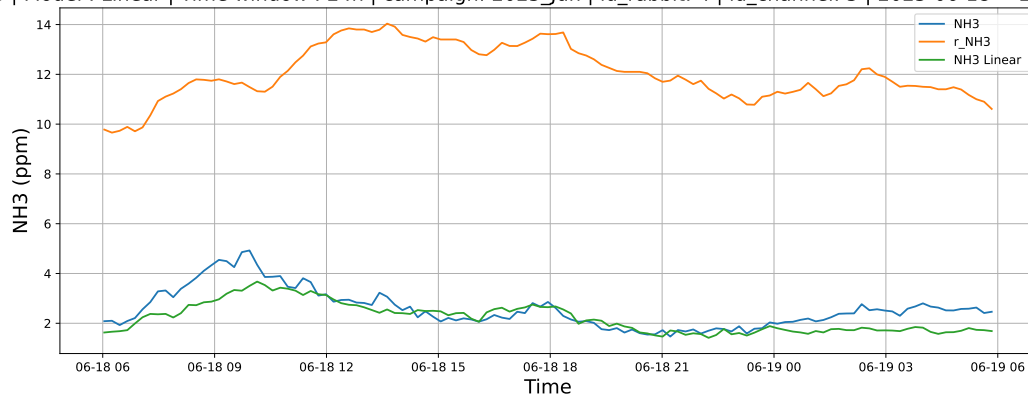
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



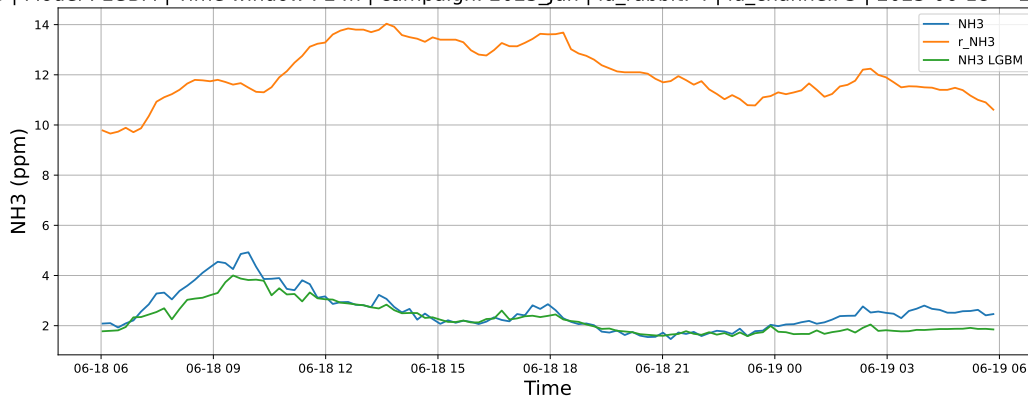
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

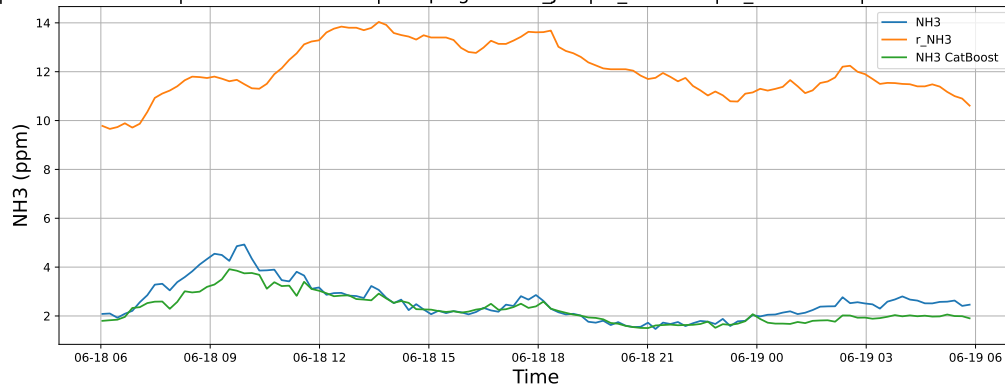
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

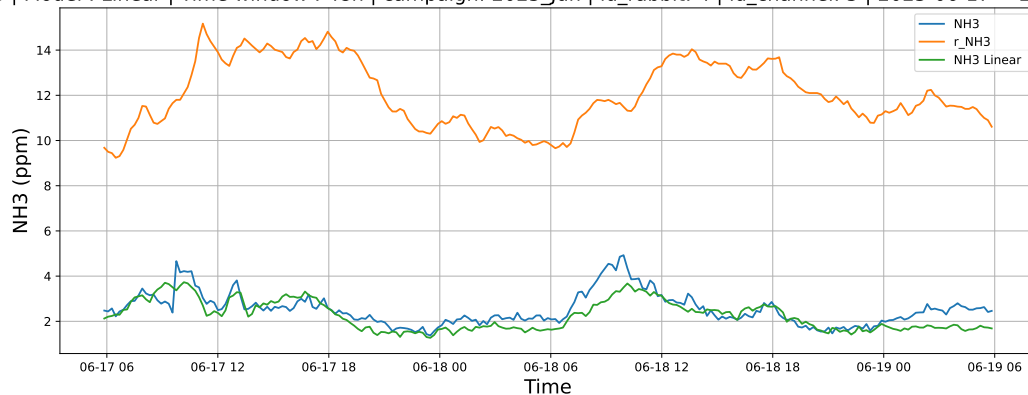


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

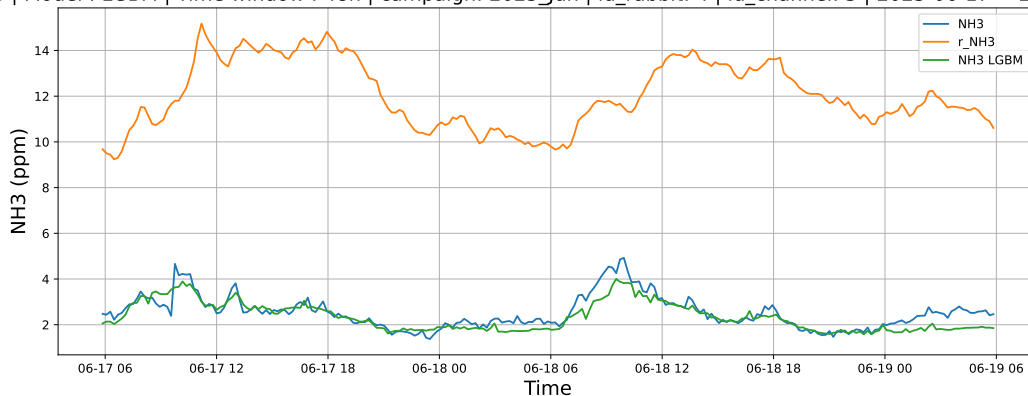


4.1.3.2 Time window : 48

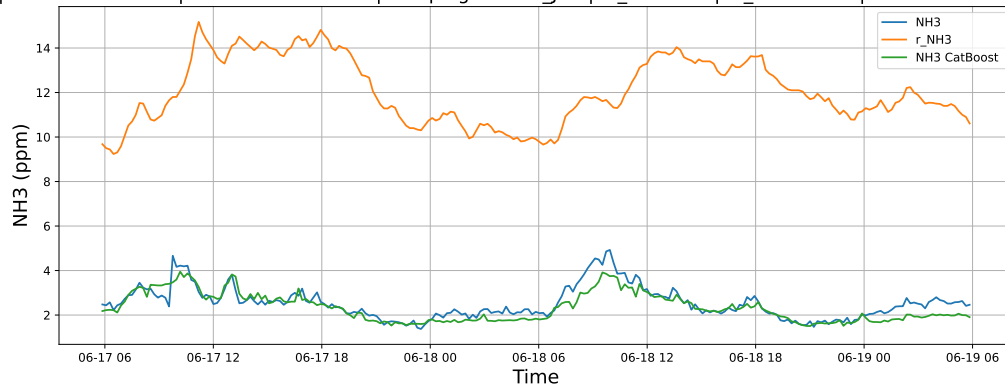
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

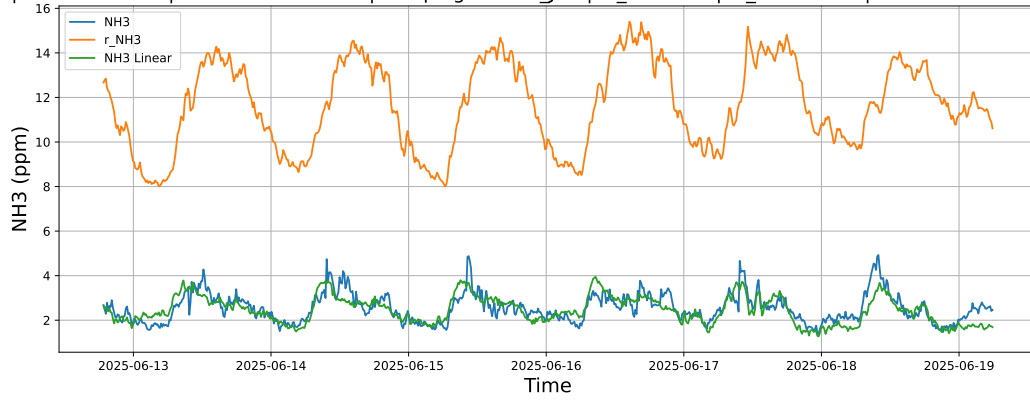


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

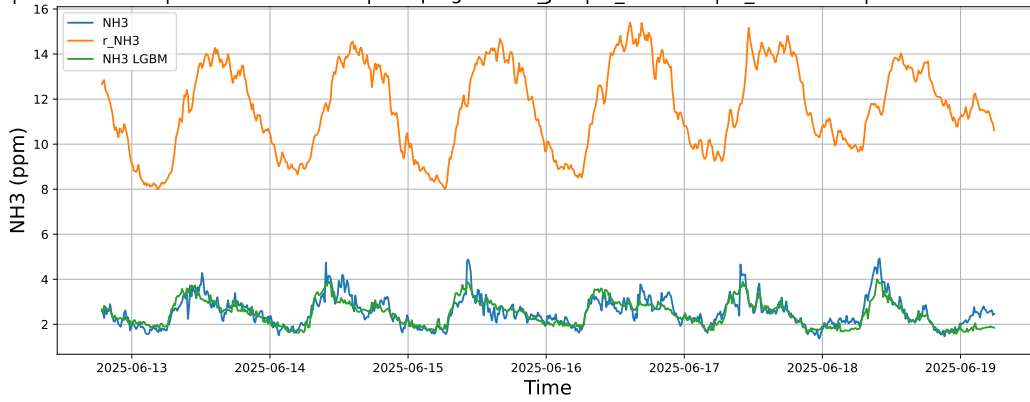


4.1.3.3 Time window : ALL

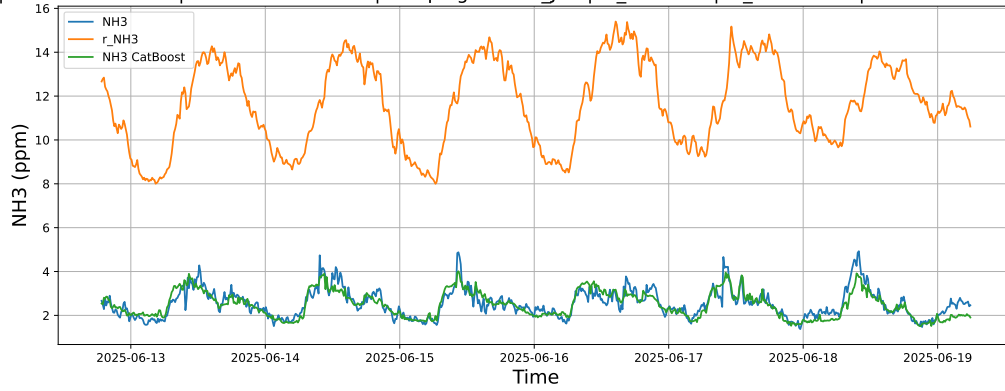
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



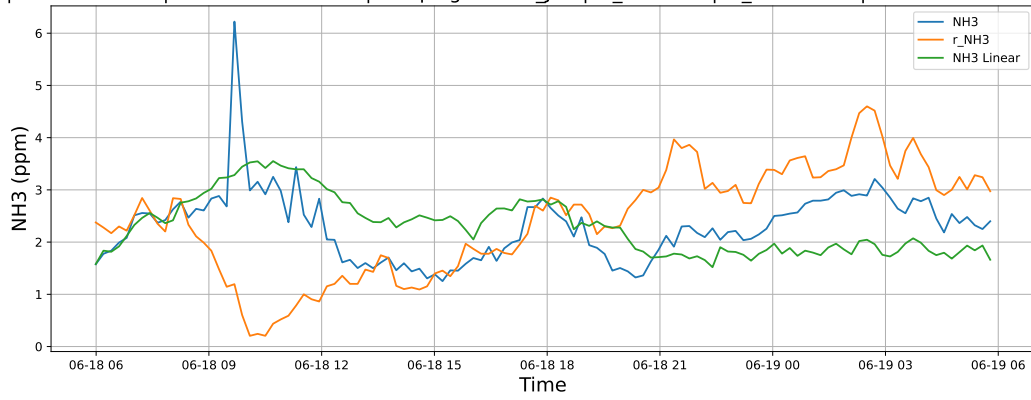
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



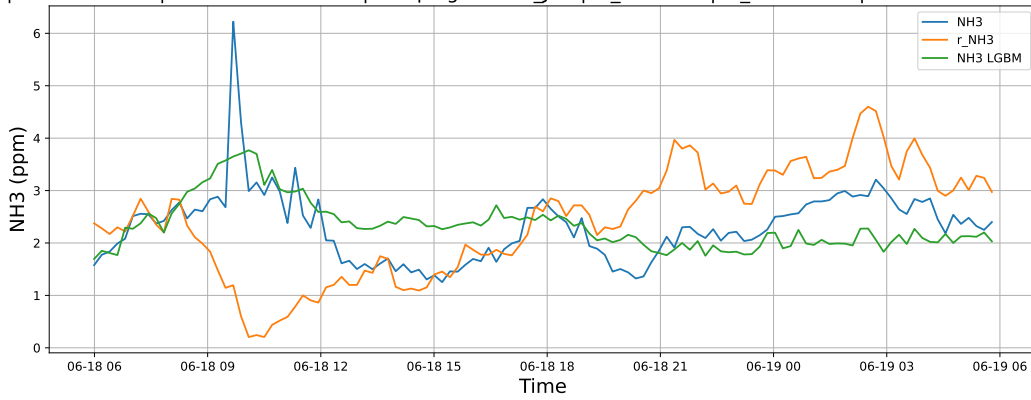
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

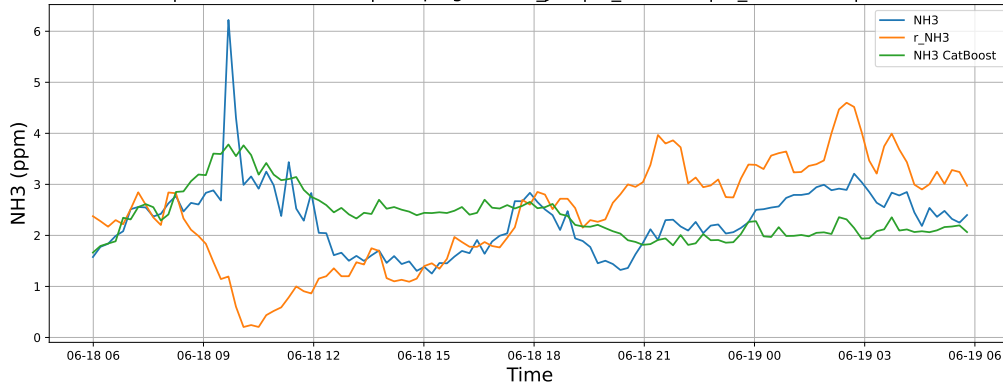
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

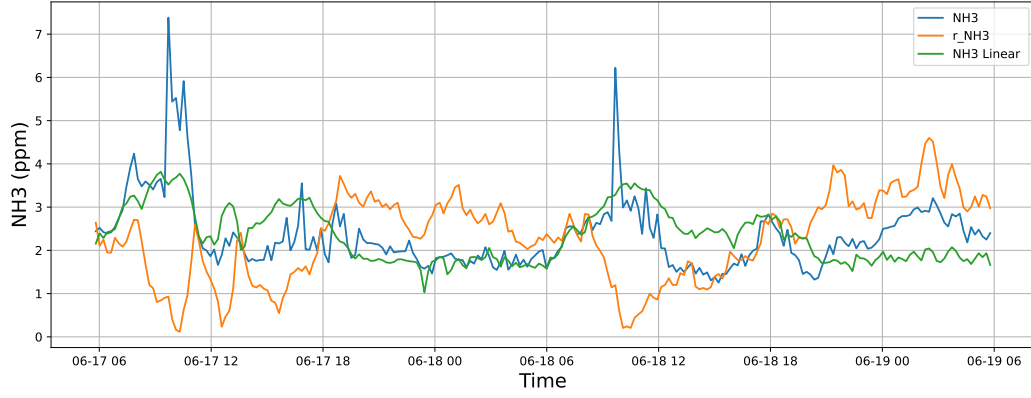


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

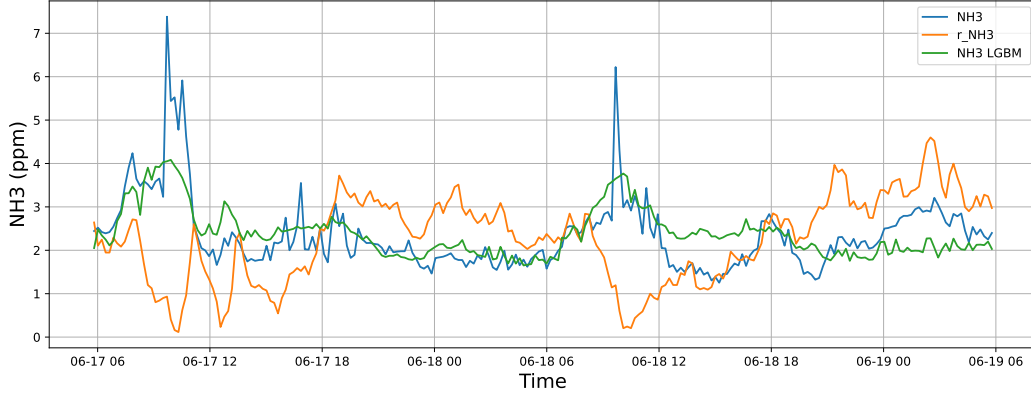


4.1.4.2 Time window : 48

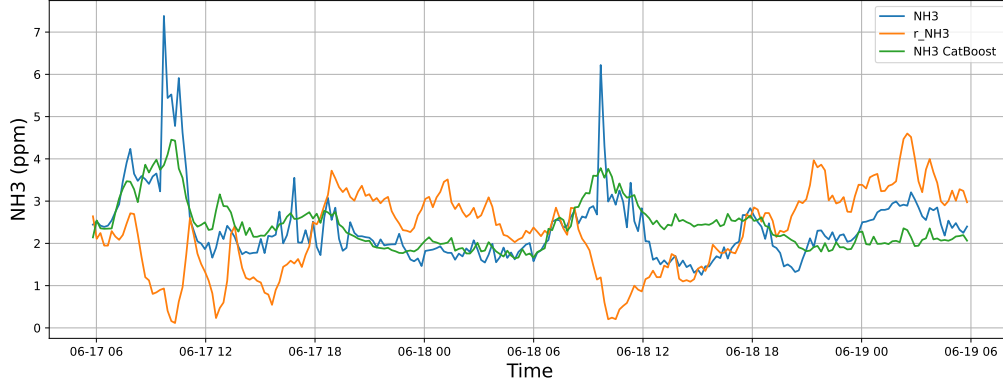
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

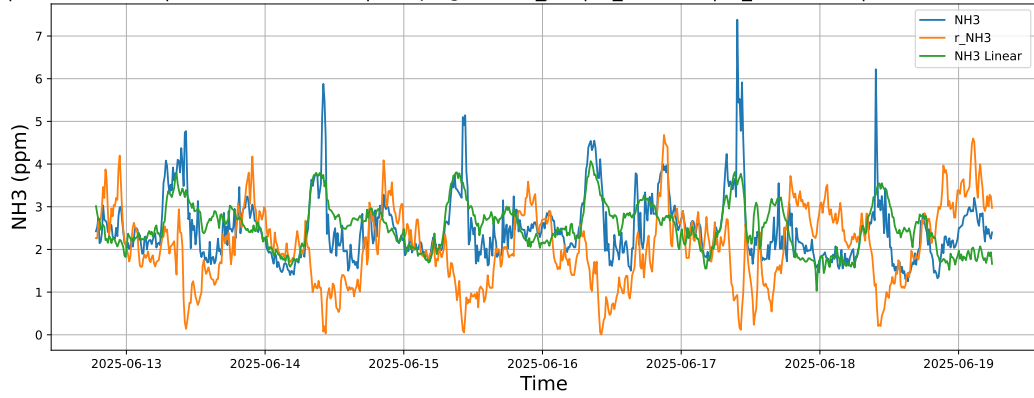


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

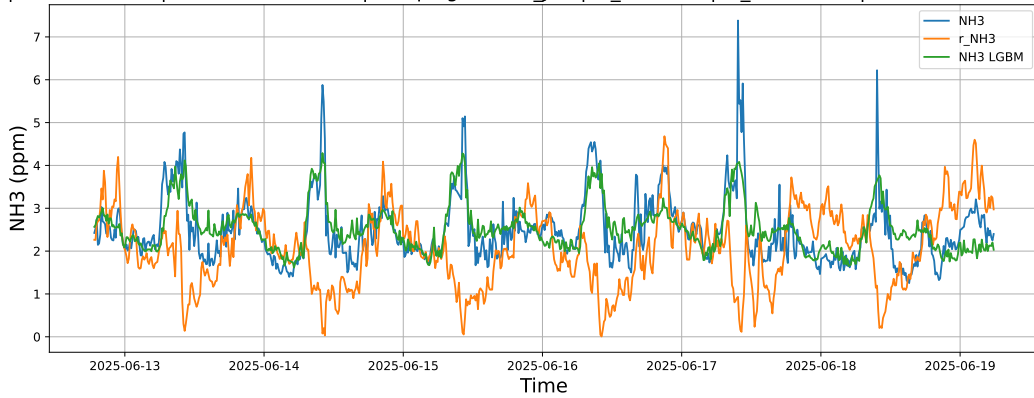


4.1.4.3 Time window : ALL

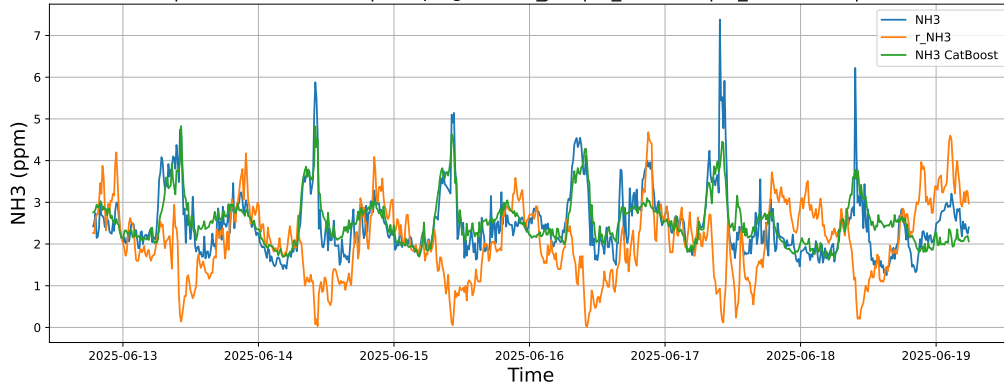
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



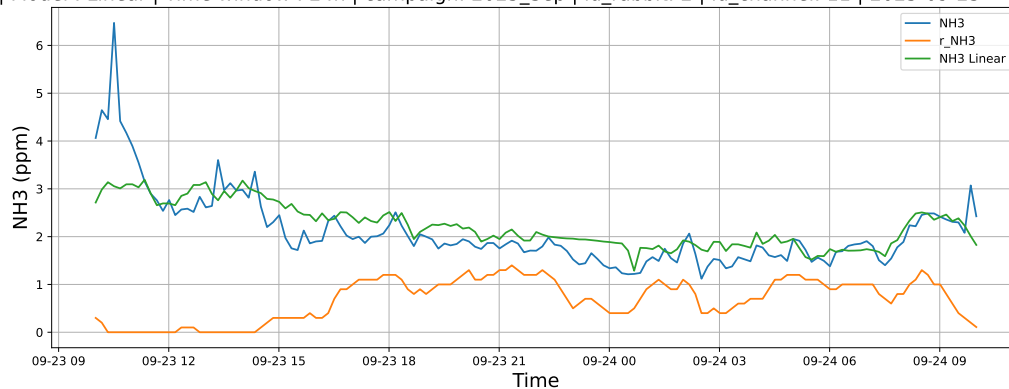
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



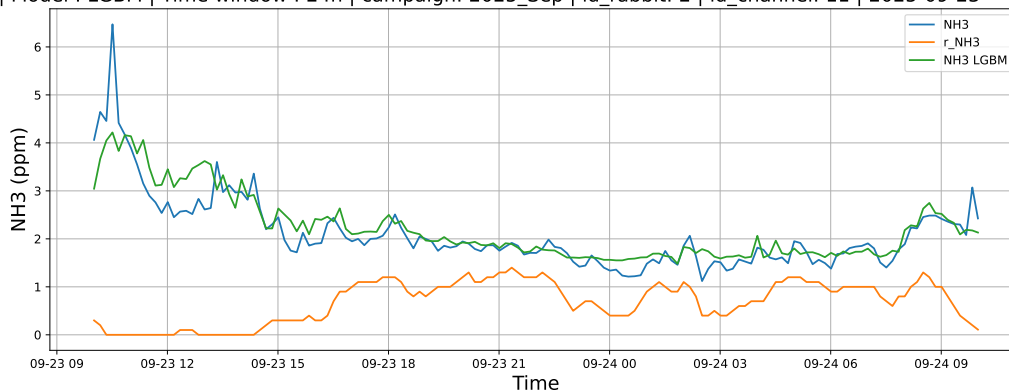
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

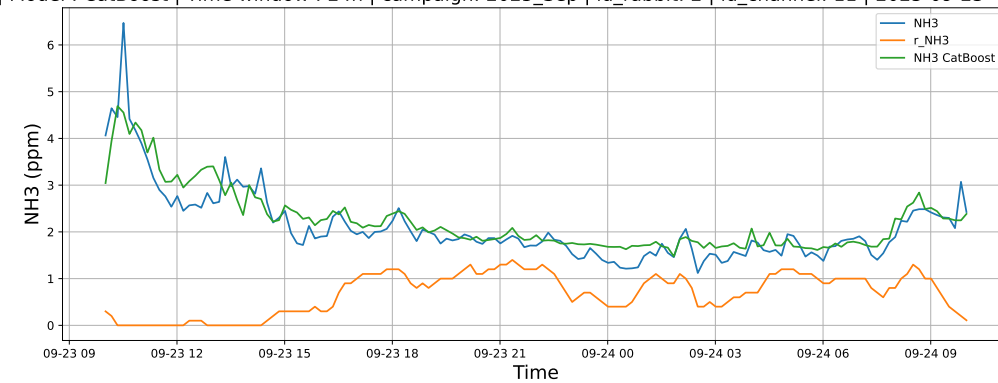
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

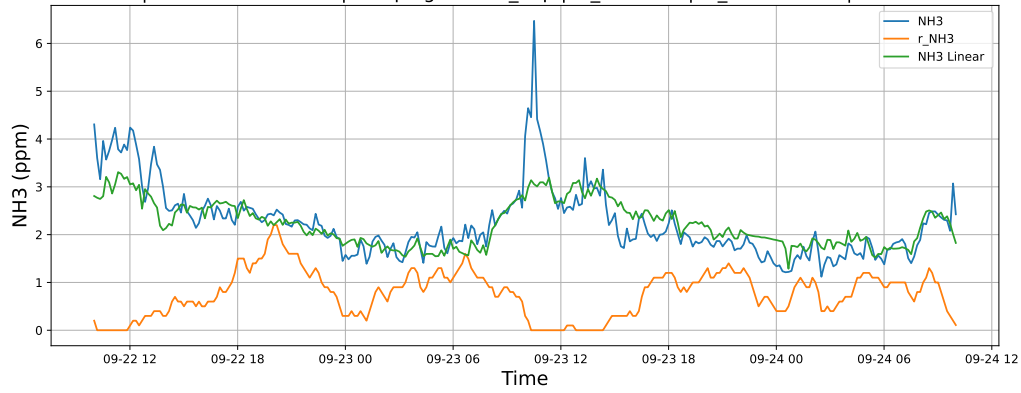


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

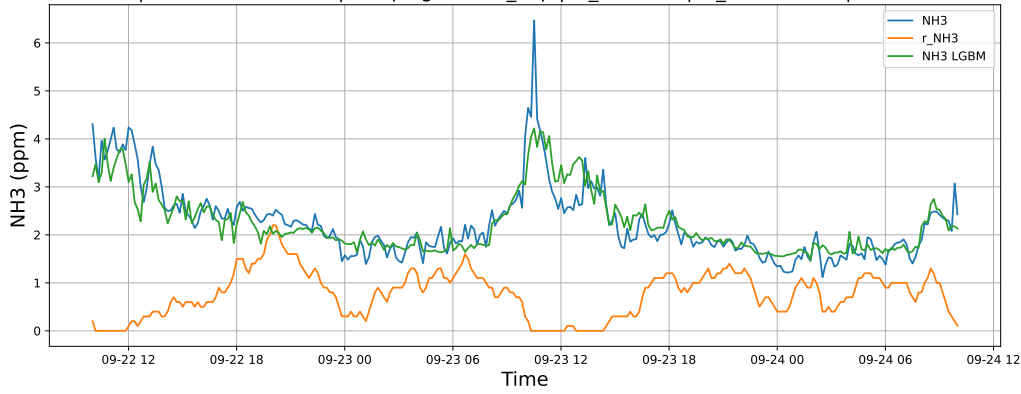


4.1.5.2 Time window : 48

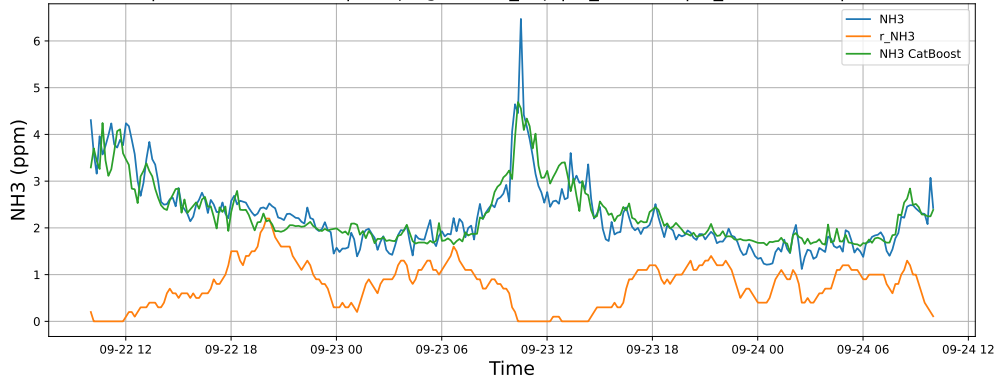
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

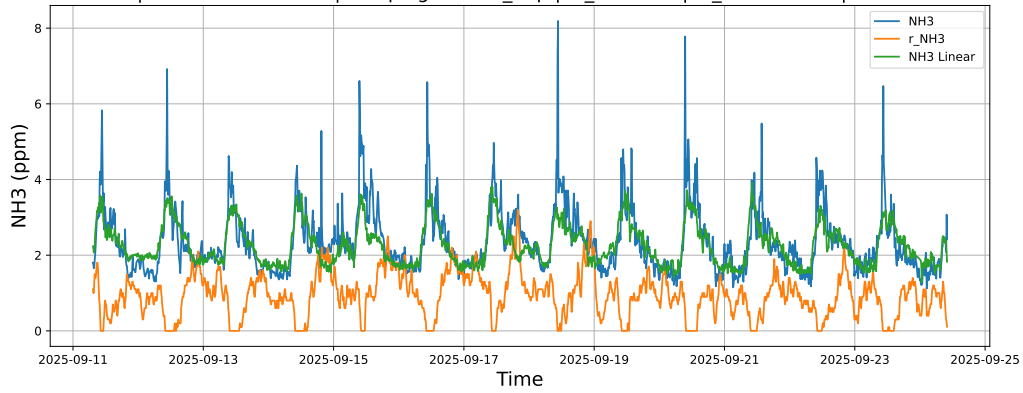


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

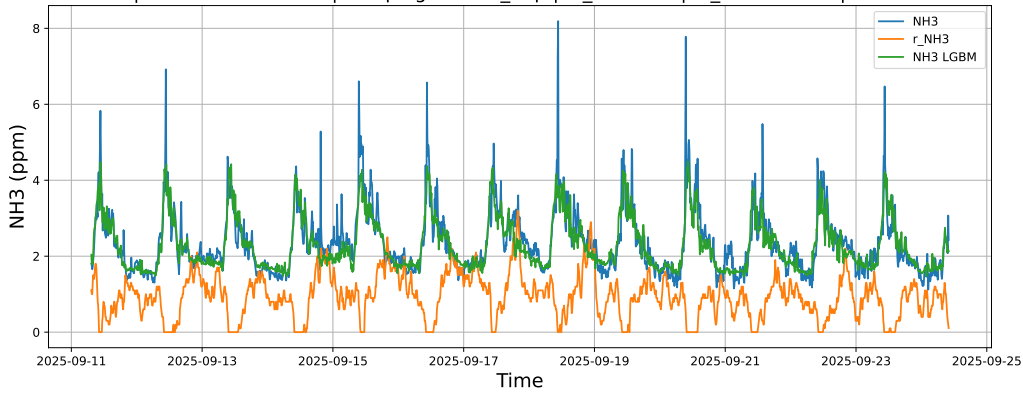


4.1.5.3 Time window : ALL

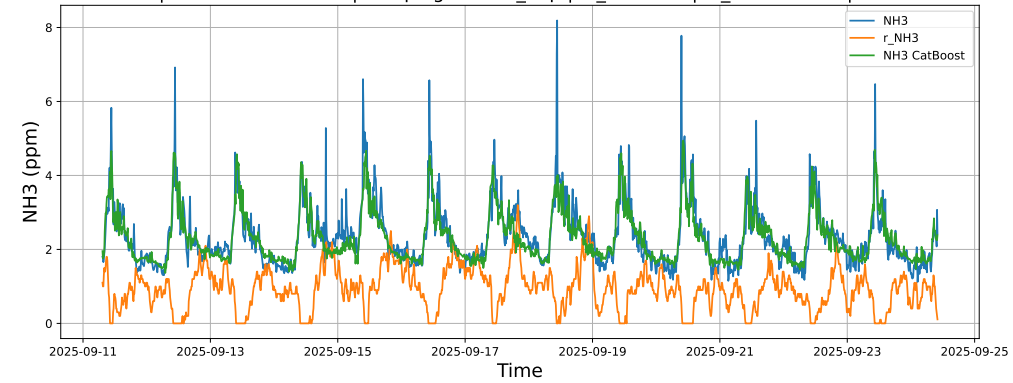
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



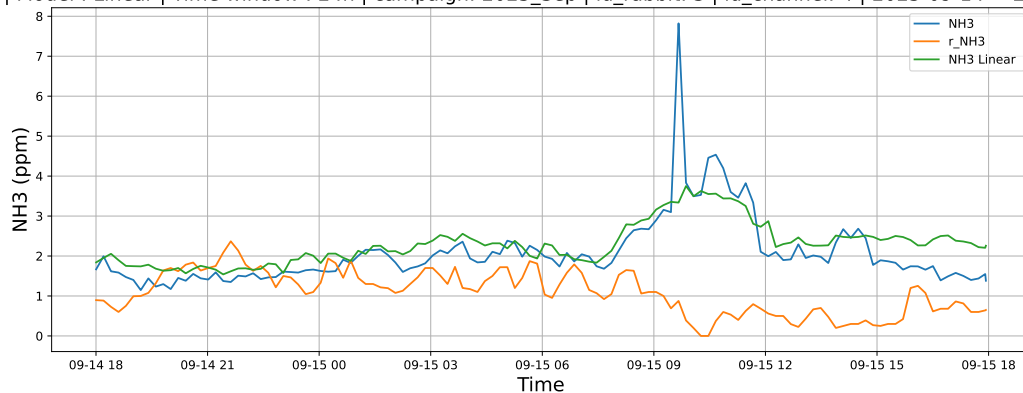
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



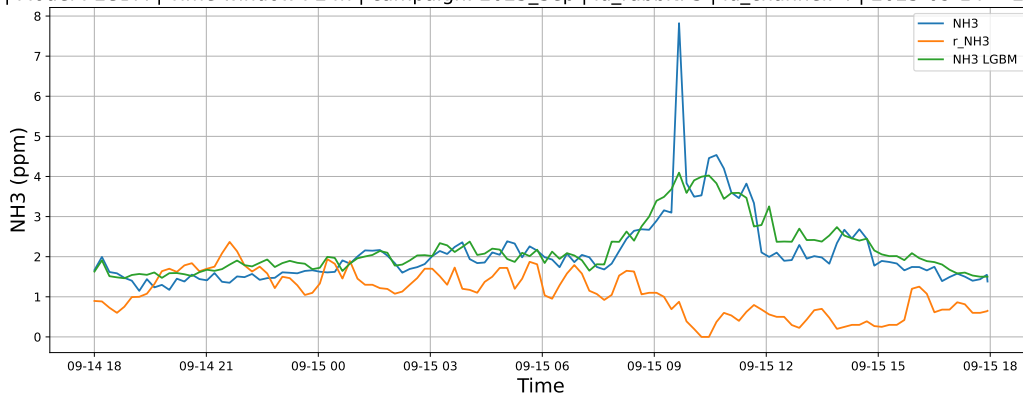
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

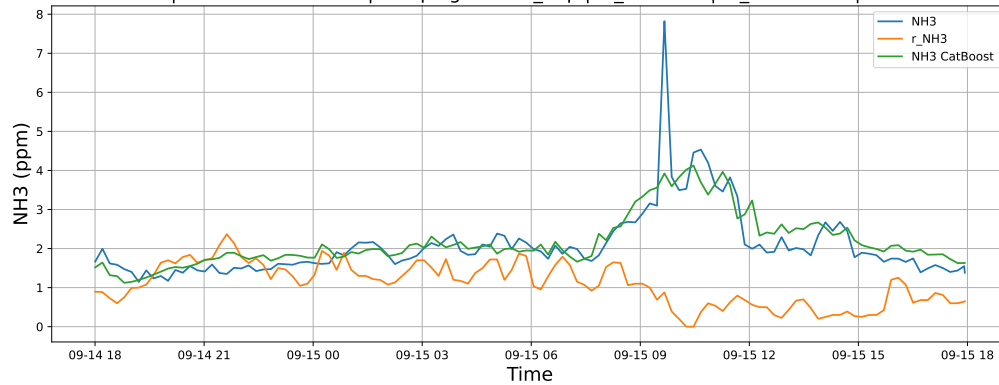
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

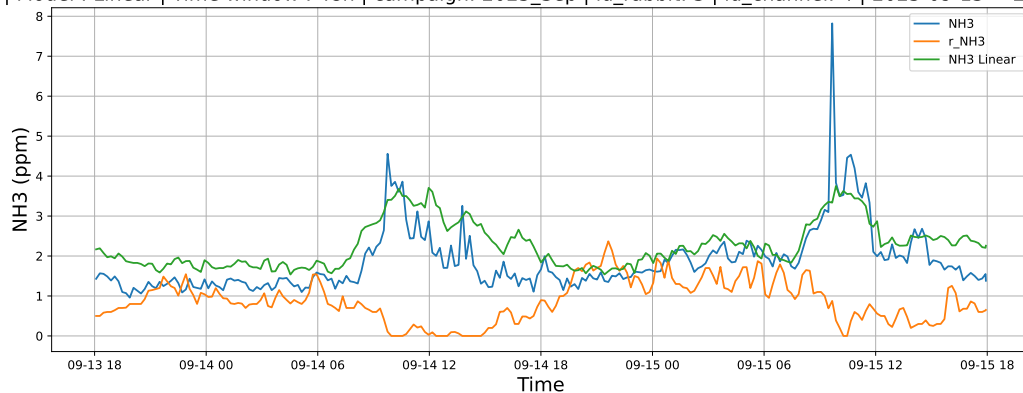


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

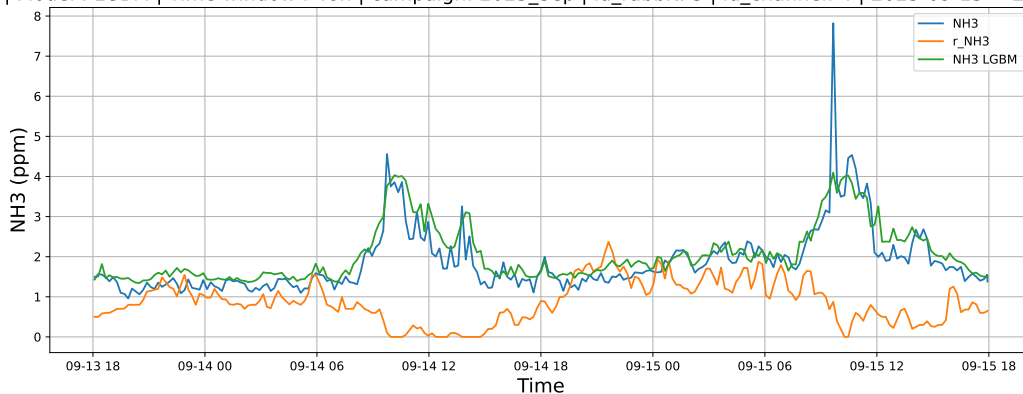


4.1.6.2 Time window : 48

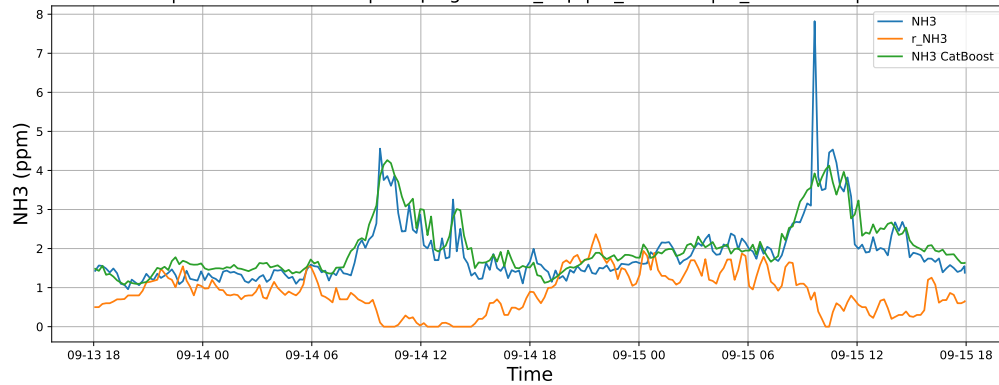
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

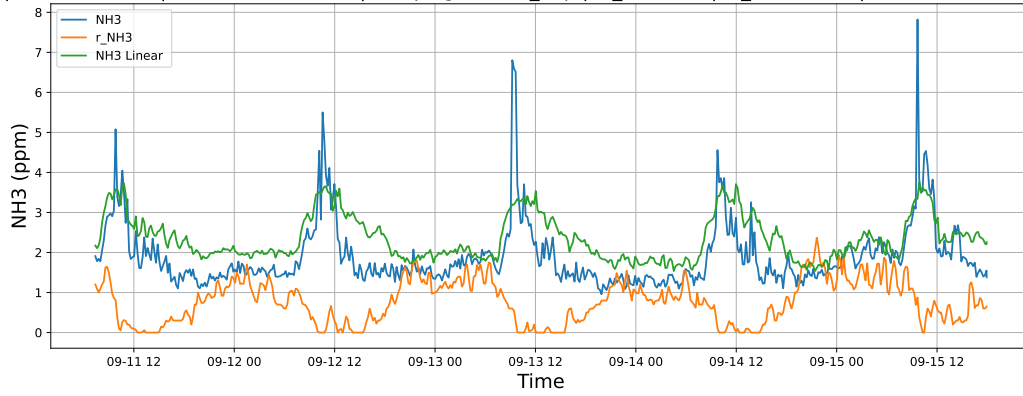


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

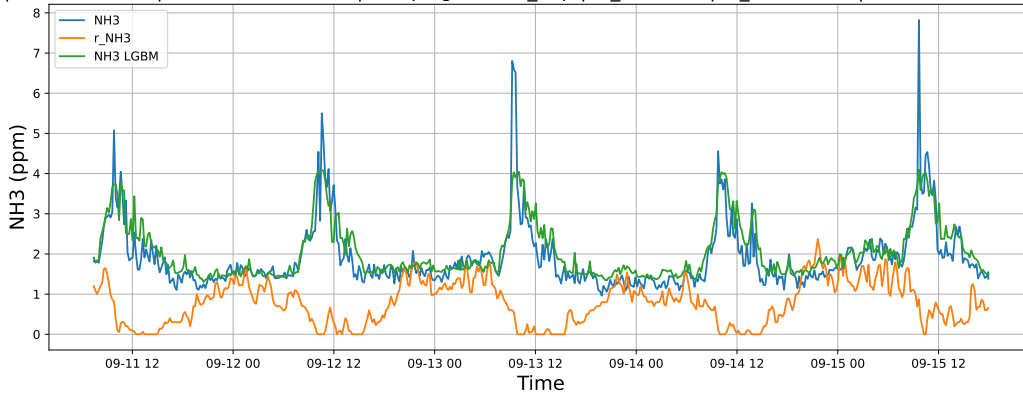


4.1.6.3 Time window : ALL

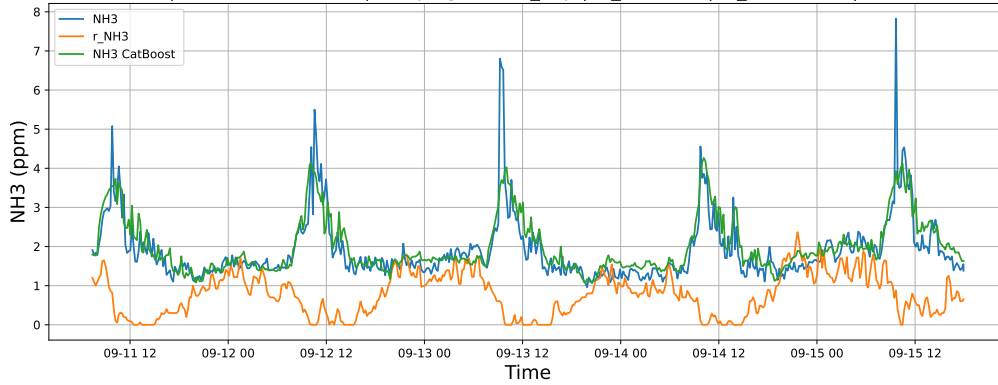
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



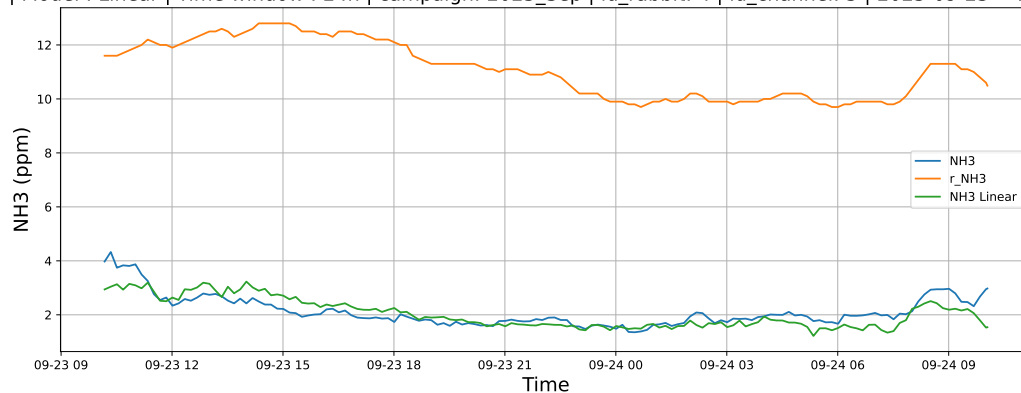
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



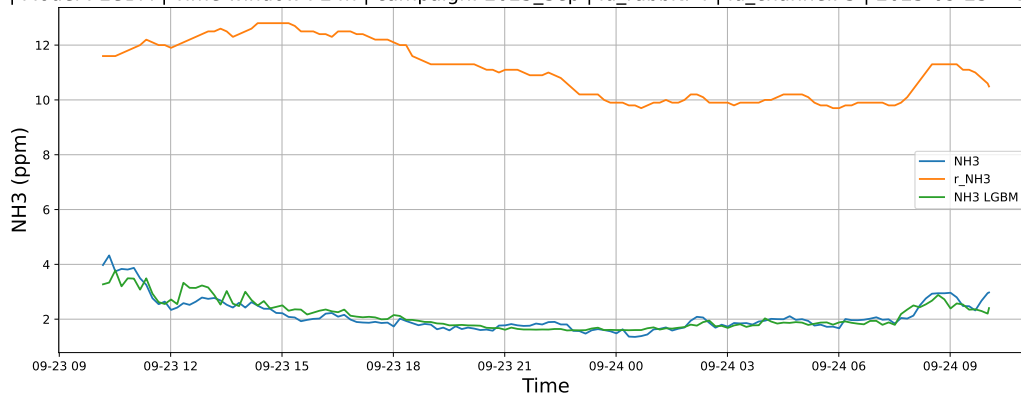
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

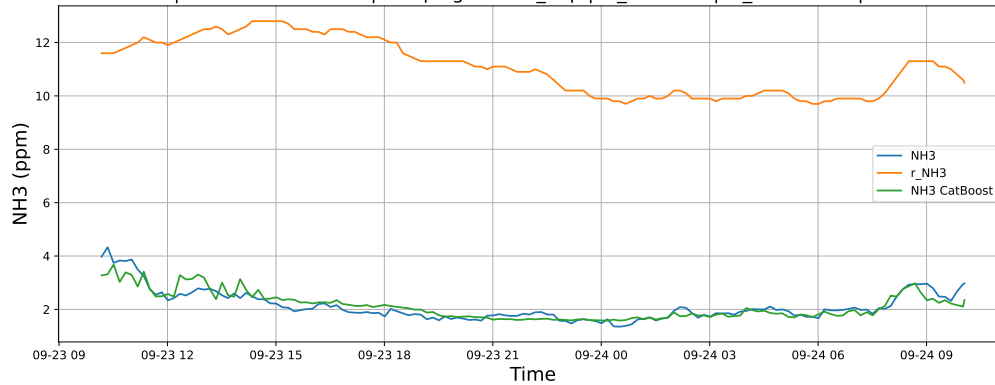
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

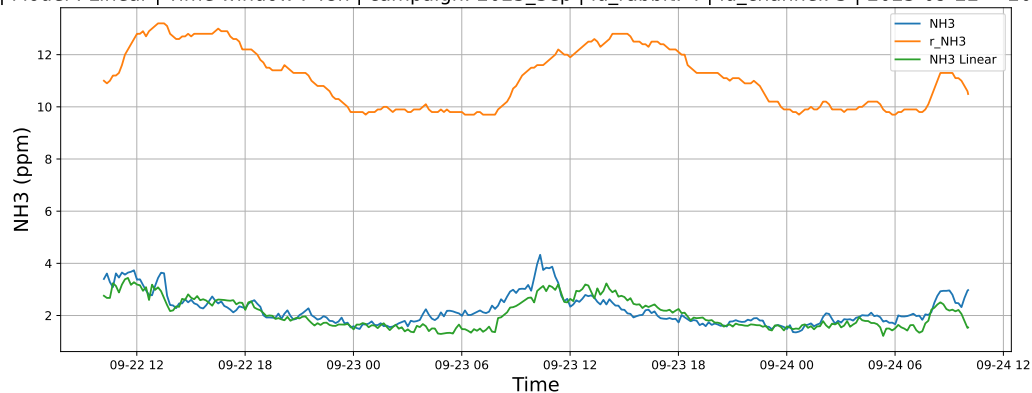


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

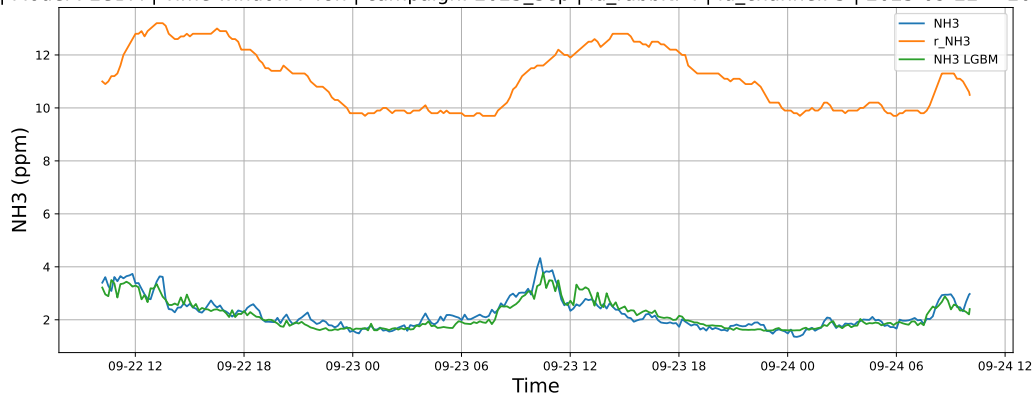


4.1.7.2 Time window : 48

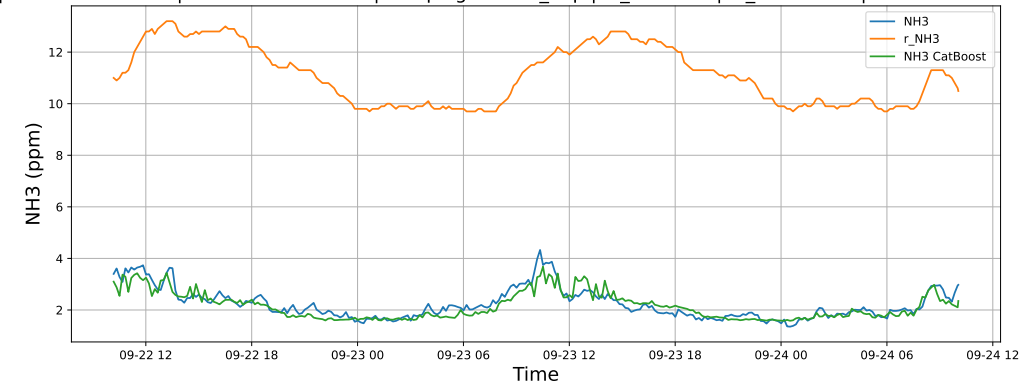
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

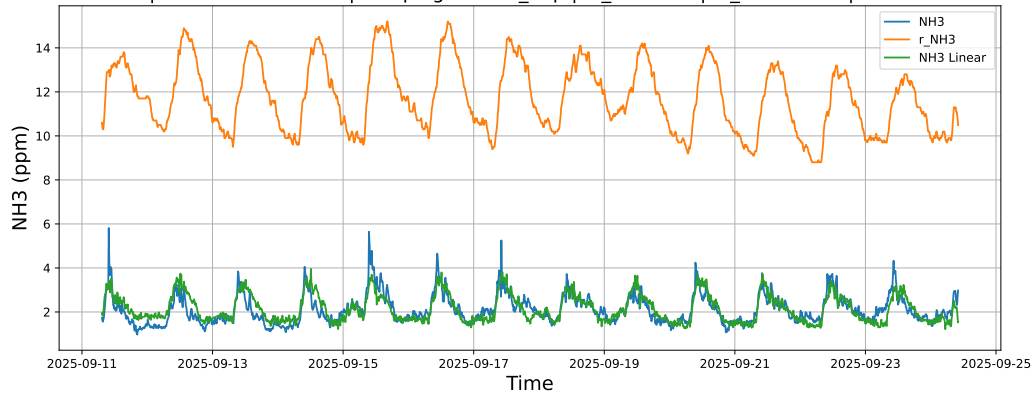


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

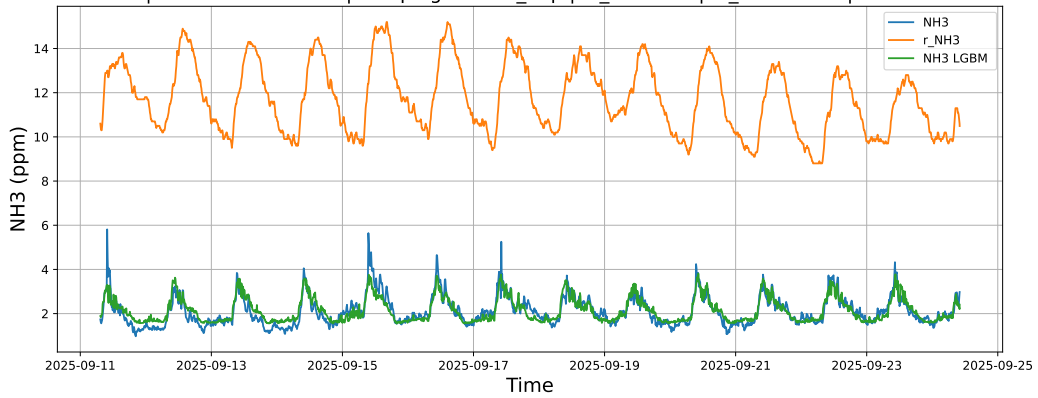


4.1.7.3 Time window : ALL

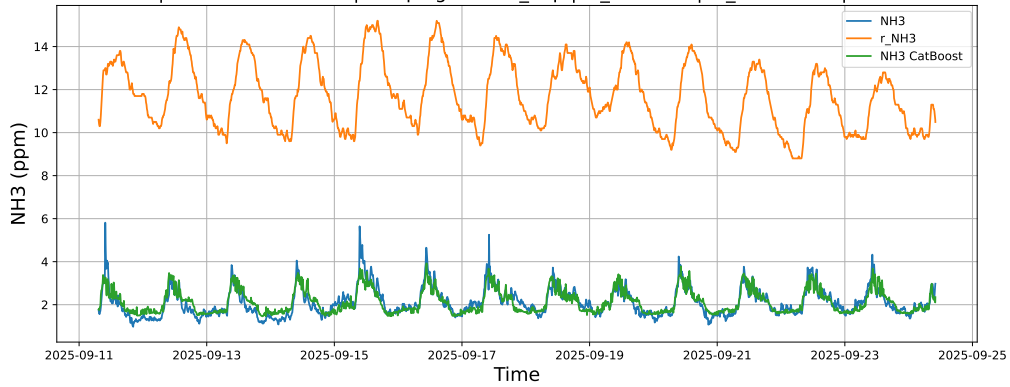
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



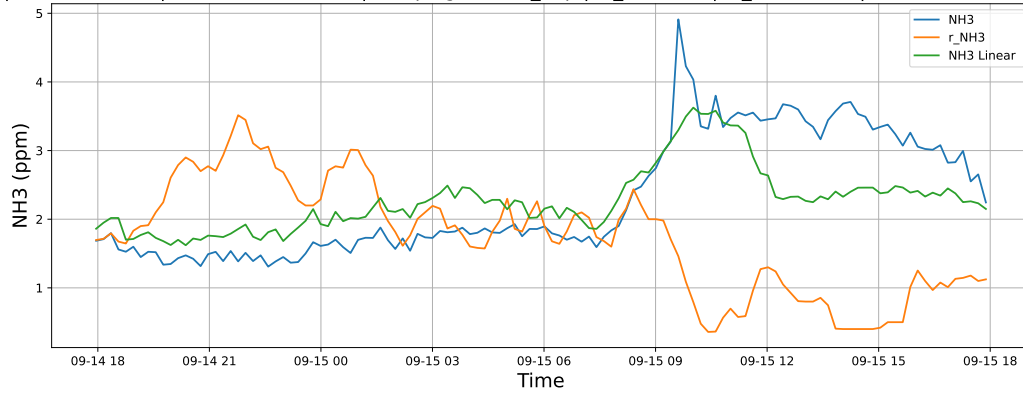
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



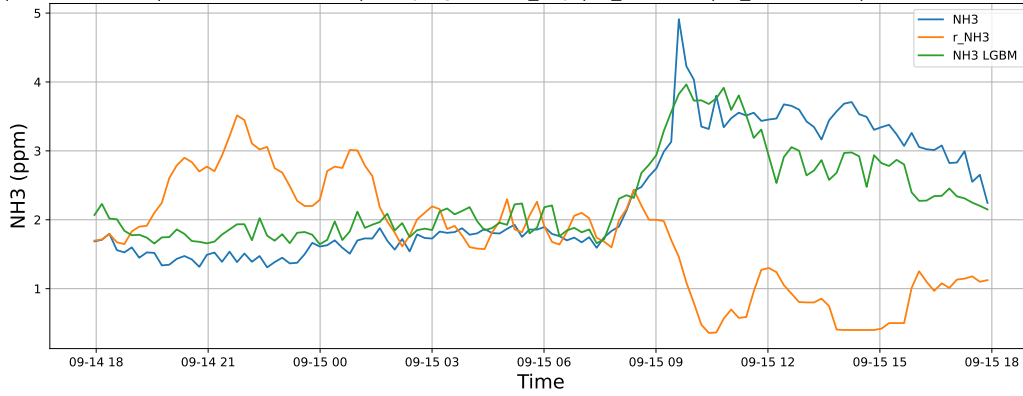
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

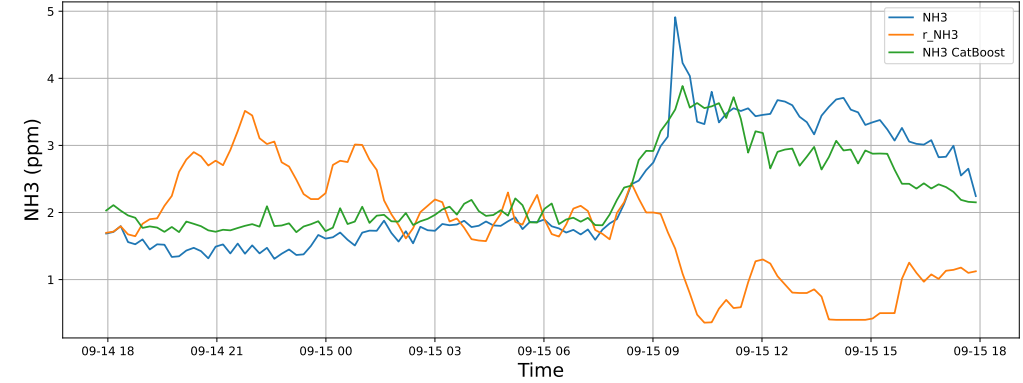
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

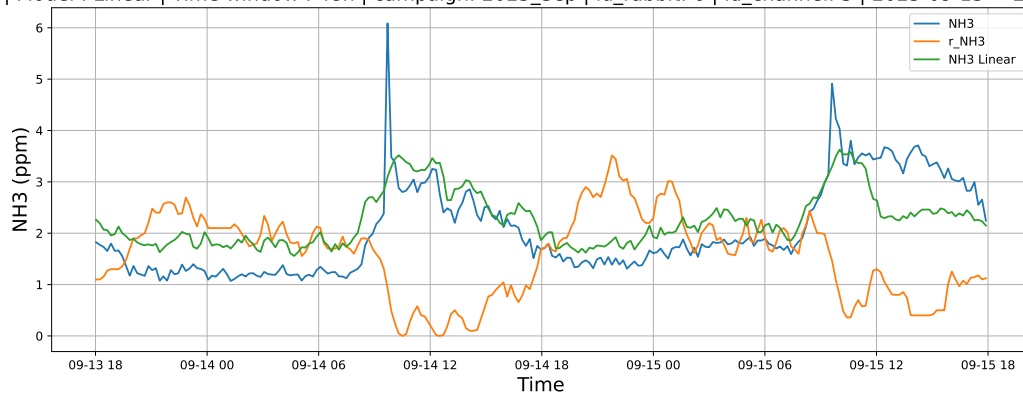


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

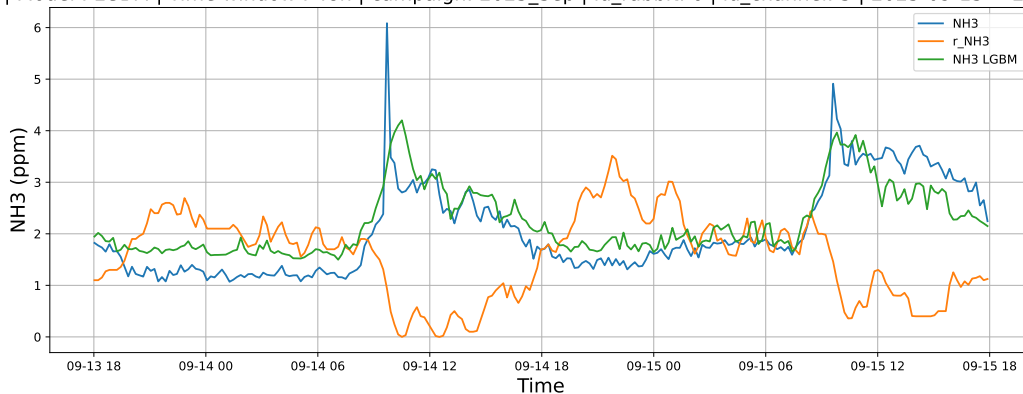


4.1.8.2 Time window : 48

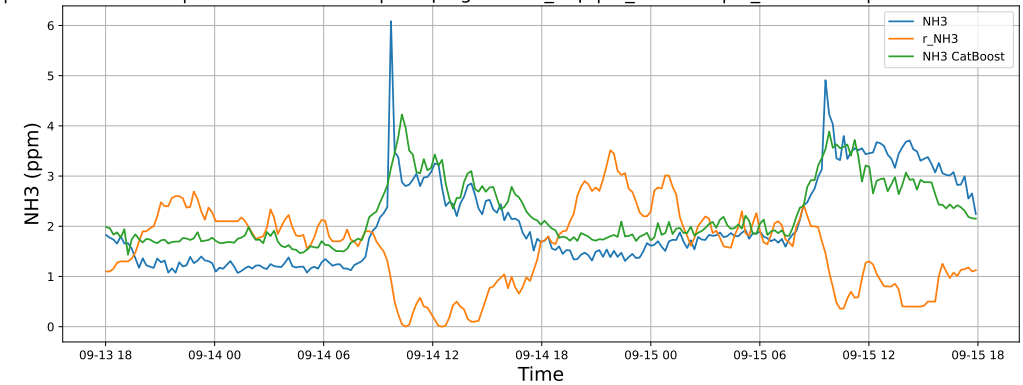
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

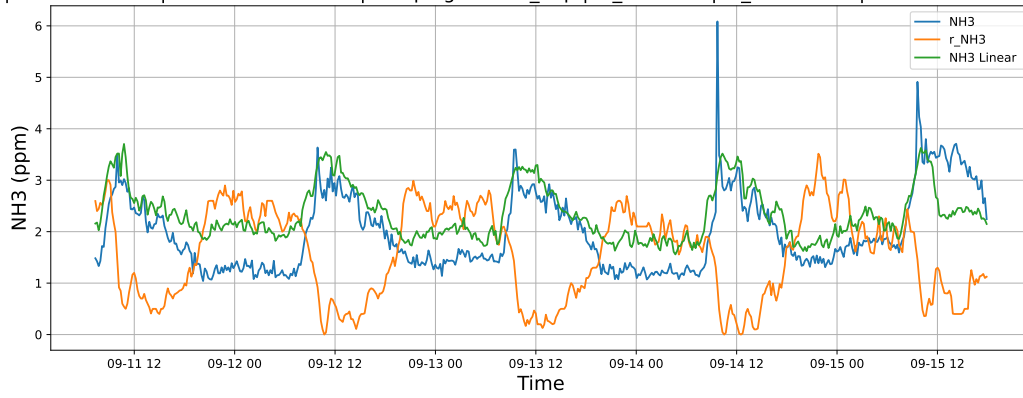


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

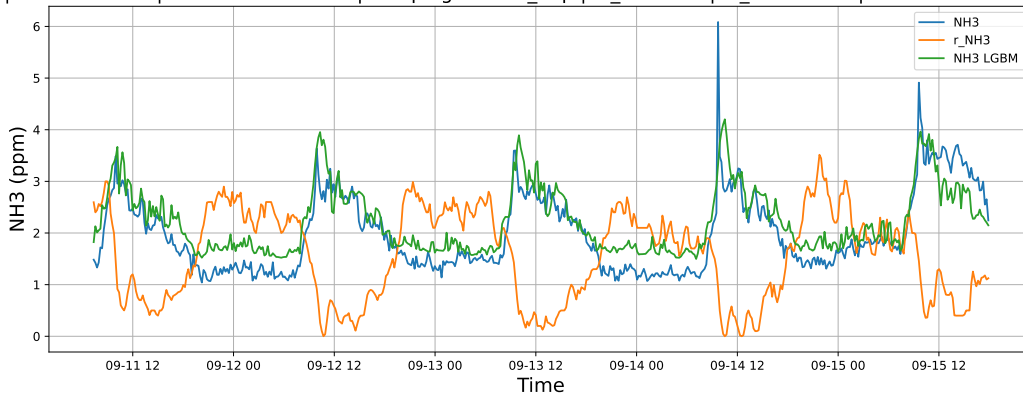


4.1.8.3 Time window : ALL

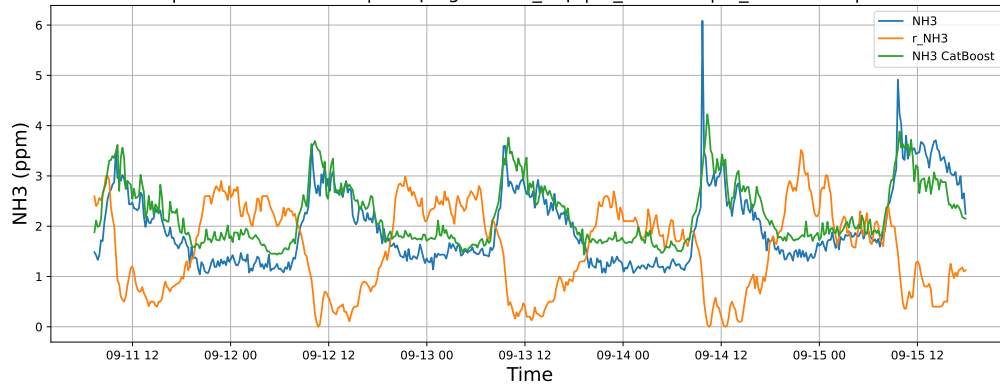
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

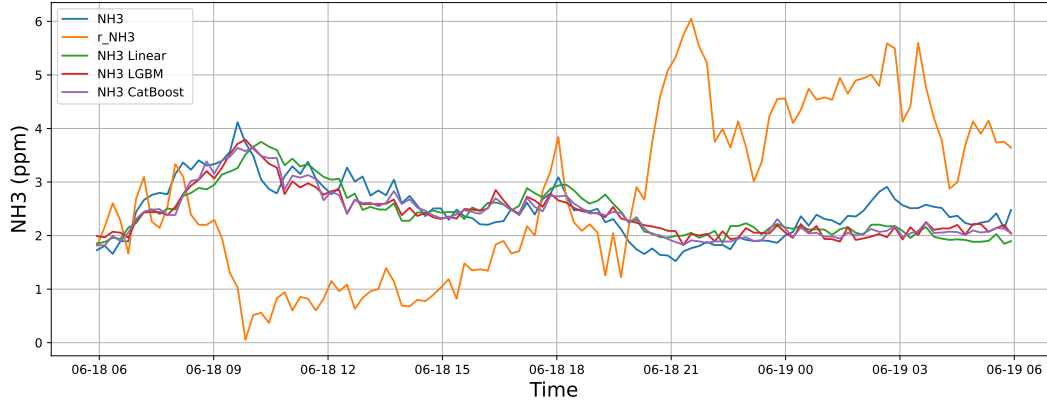


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

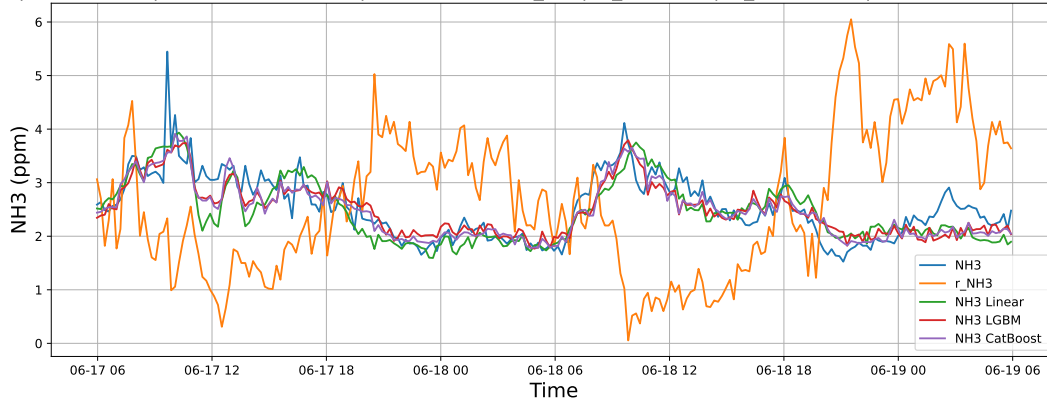
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



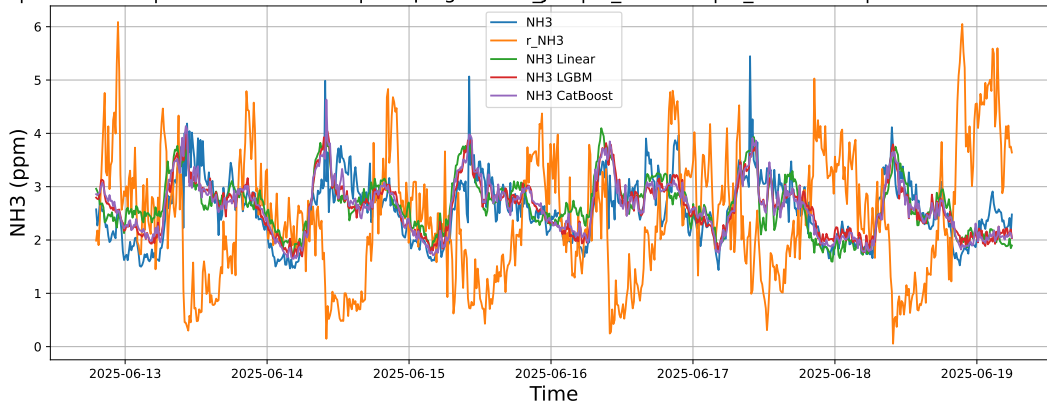
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

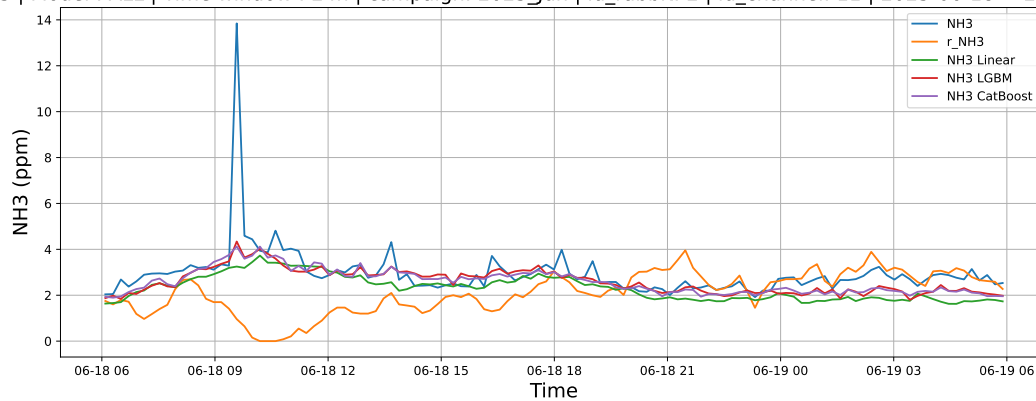
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

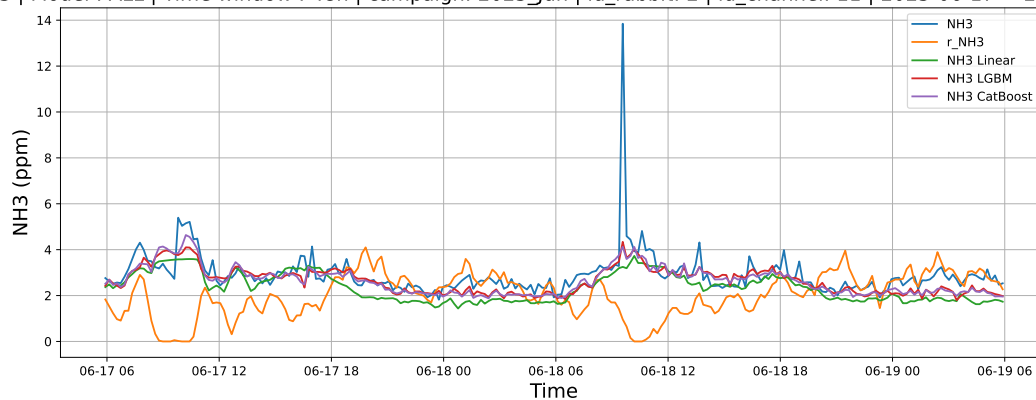
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



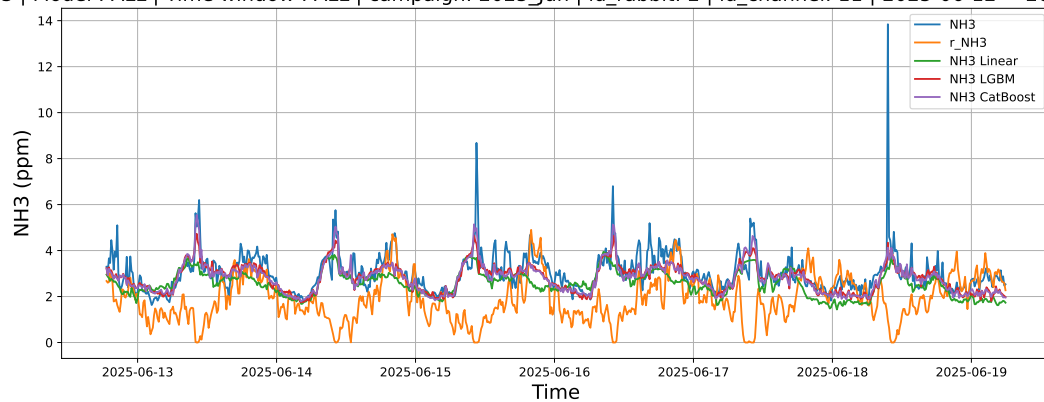
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

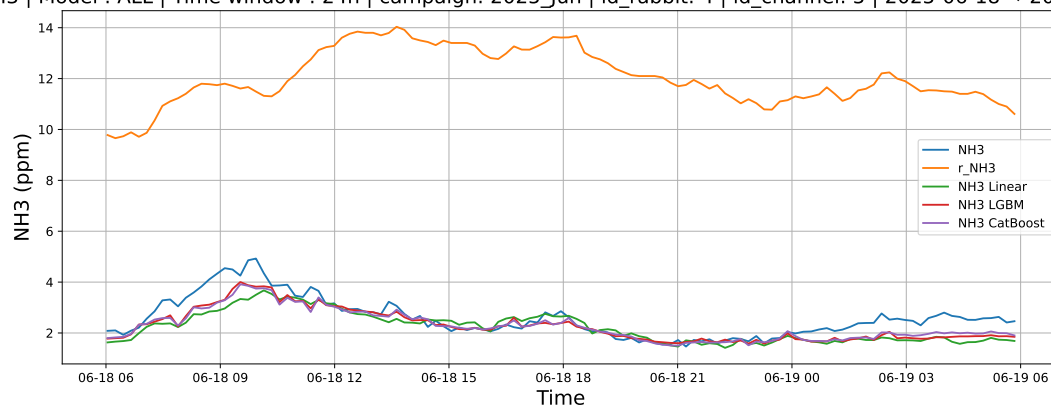
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

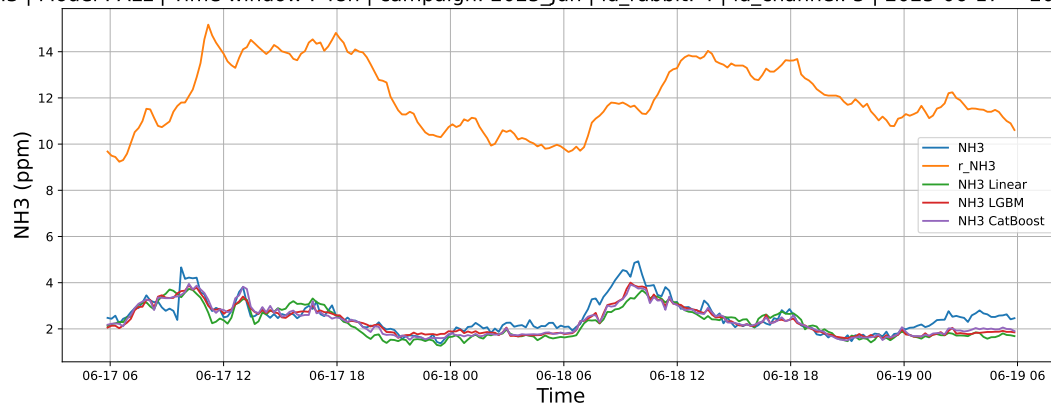
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



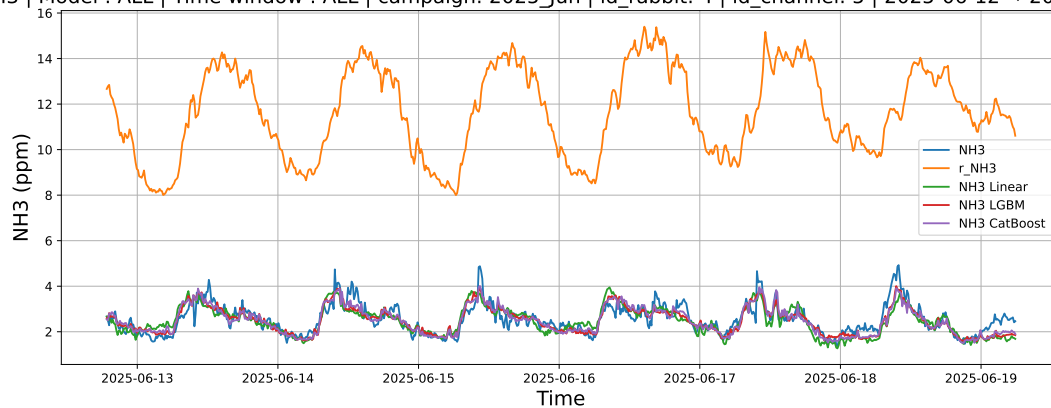
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

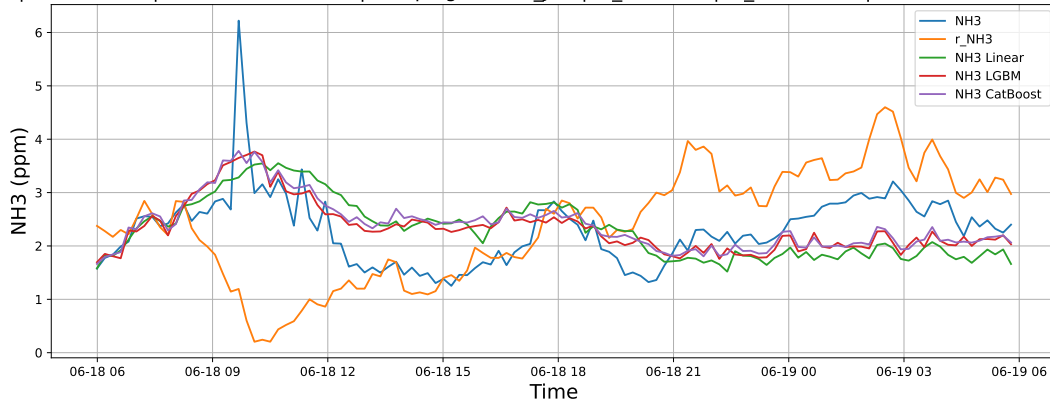
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

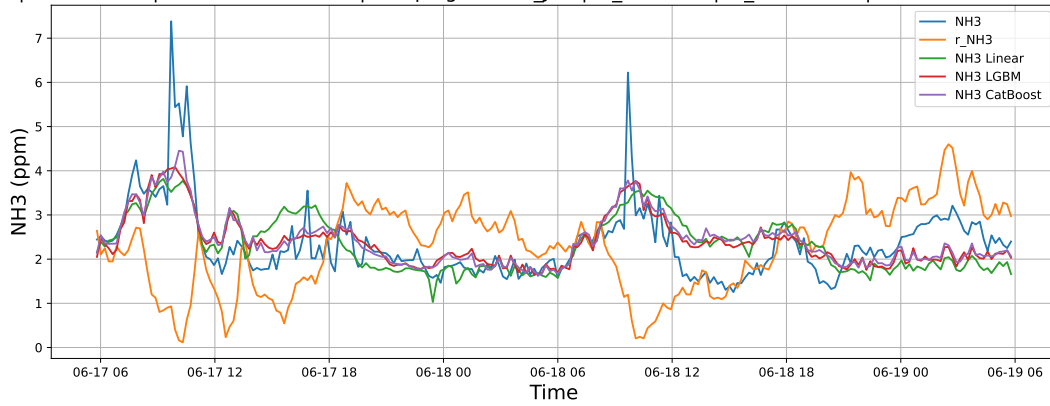
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



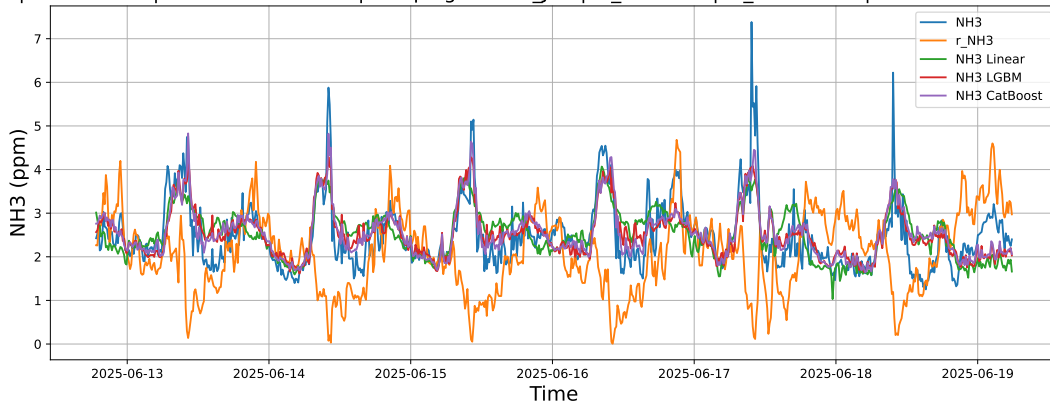
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

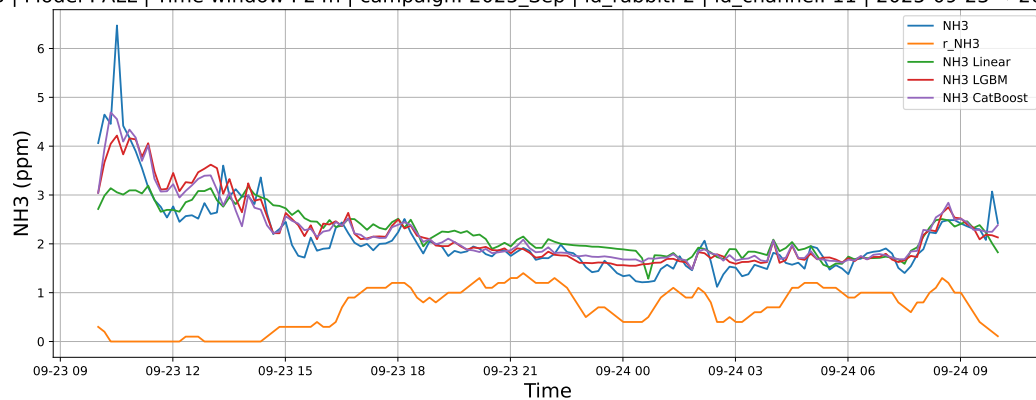
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

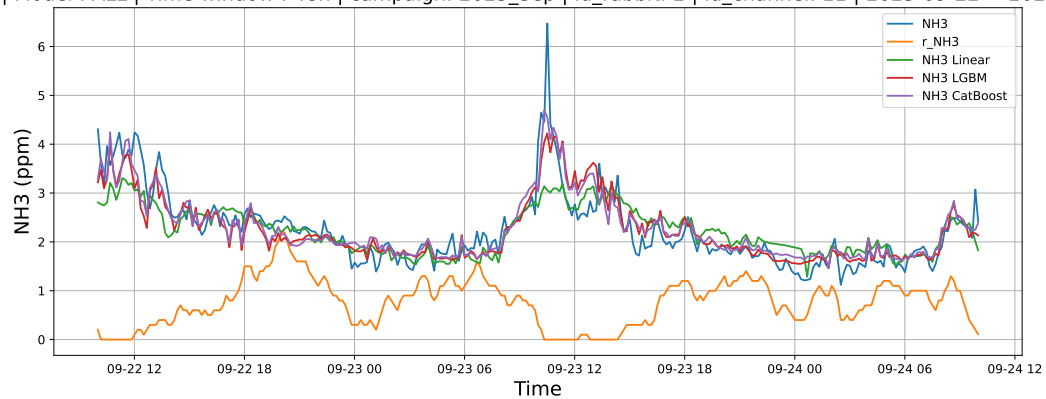
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



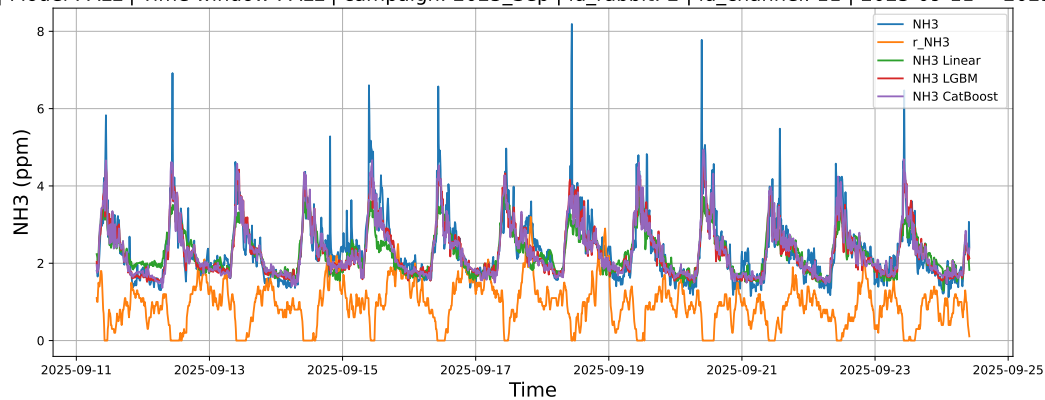
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

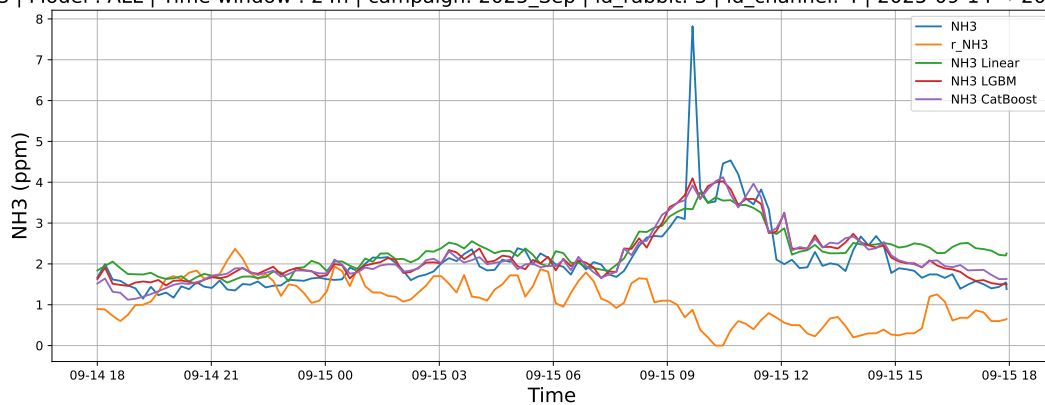
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

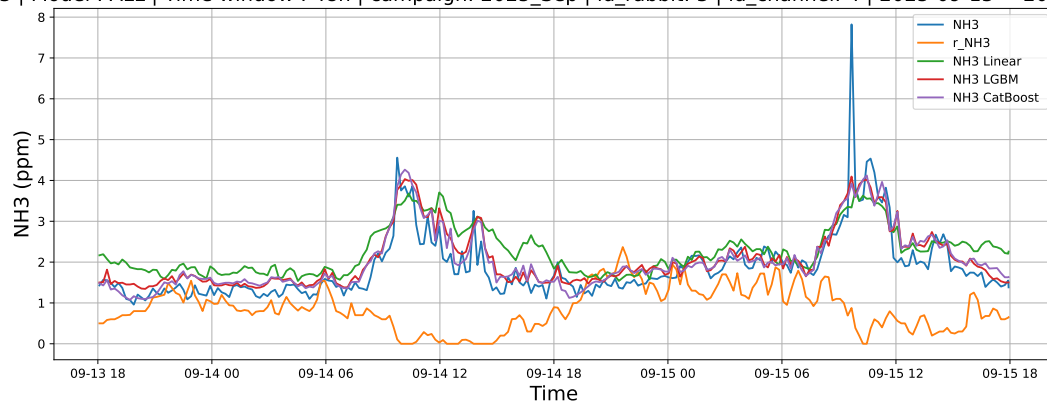
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



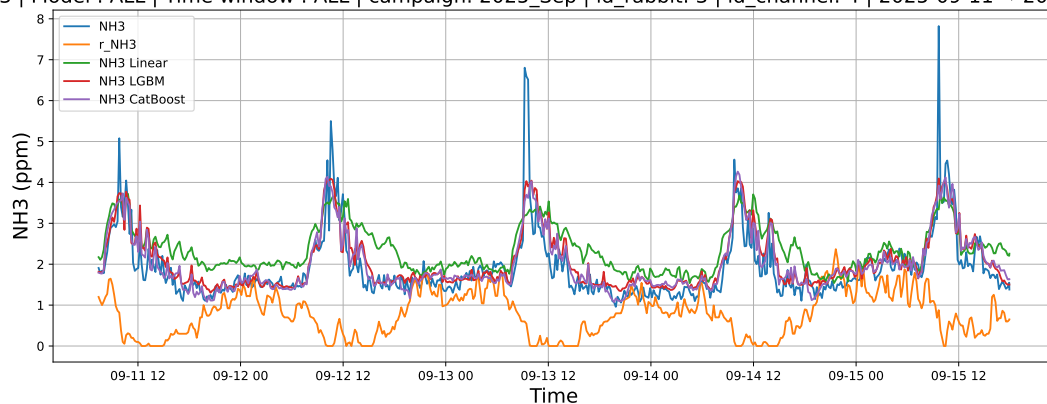
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

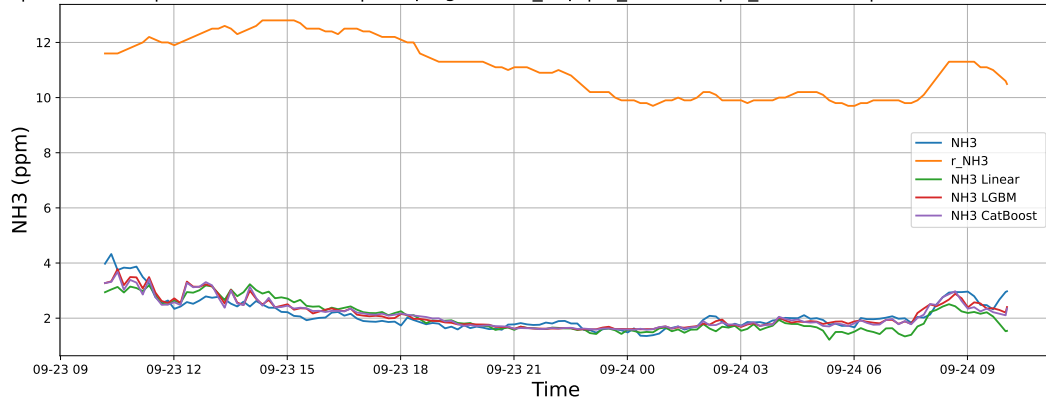
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

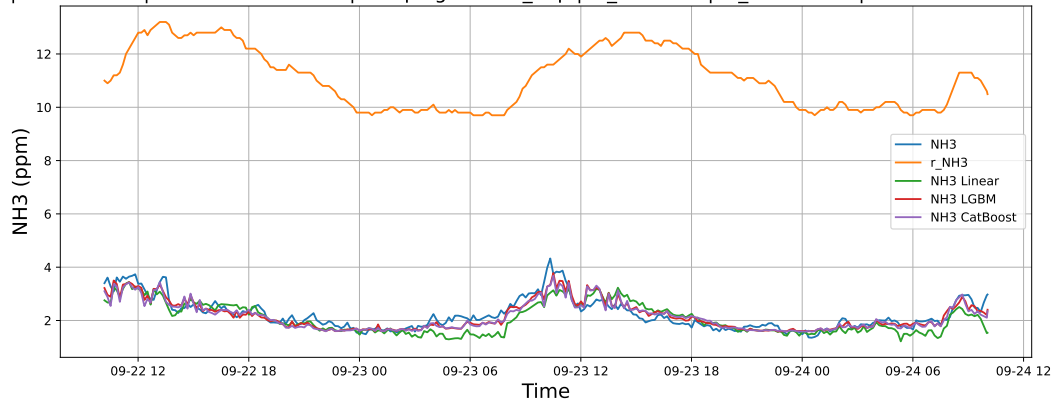
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



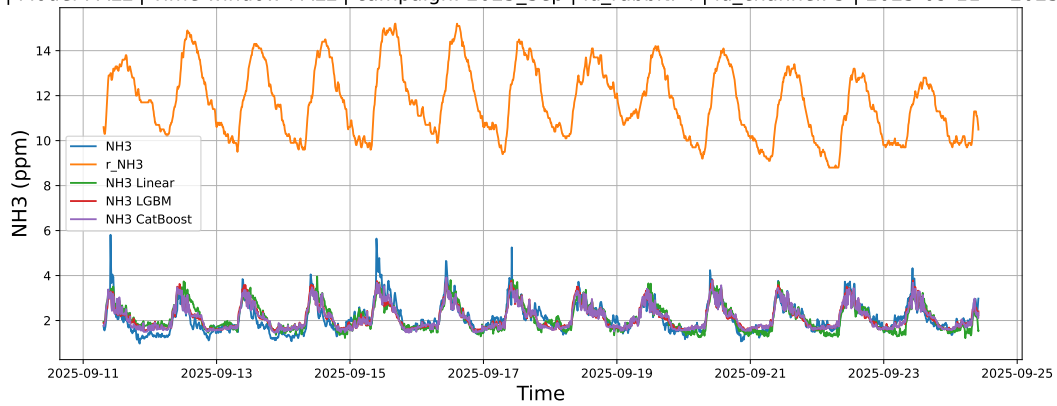
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

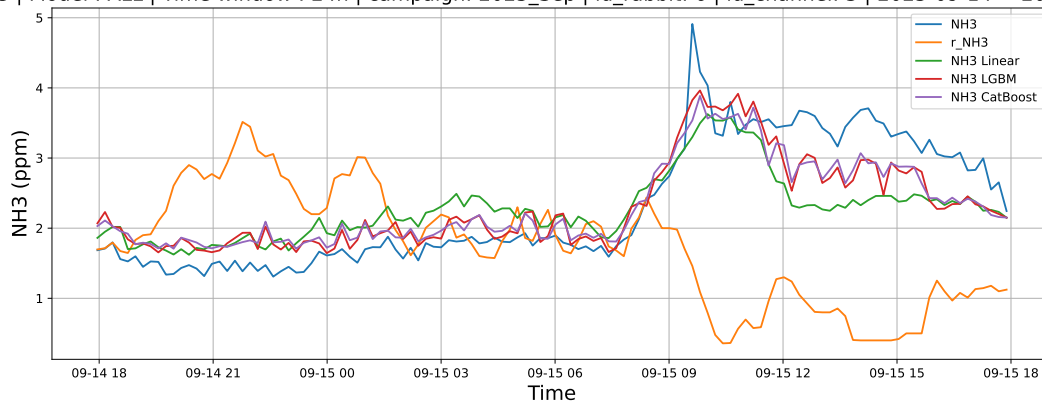
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

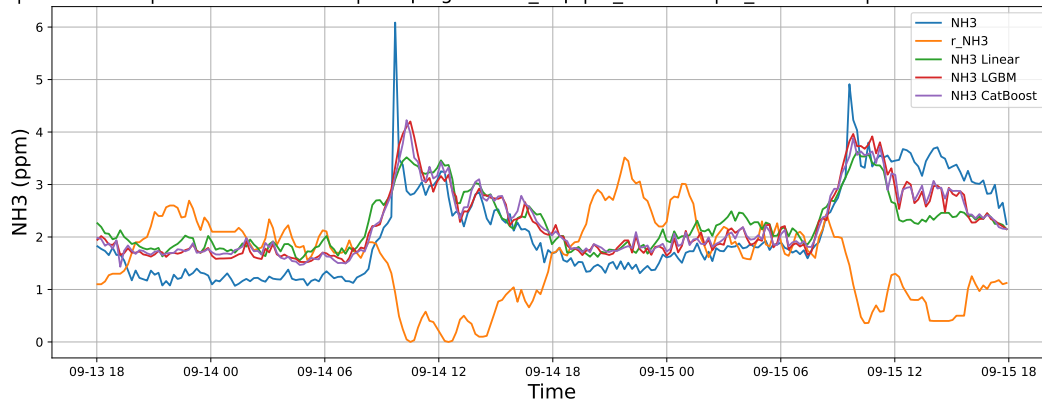
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

